TriLens-MoE-Stack: A Multi-View Hybrid Framework for Lineage-Aware Multi-Drug Antibiotic Resistance Prediction in *Escherichia coli* Using Genomic Data

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***Abstract*—Antibiotic resistance prediction using bacterial genomes is more rapid than laboratory analysis, though high scores do not indicate the performance of a model on unfamiliar bacteria. Resistance arises either from mutations in existing genes or from acquired genes, one representation cannot capture both, and existing multi-view methods combine views with fixed weights. We therefore developed TriLens-MoE-Stack. Three neural lenses read 535,439 single-nucleotide polymorphisms grouped by gene, a 32,441-gene presence profile, and four summary statistics per gene; a mixture-of-experts gate weights them per strain and antibiotic; and five classical learners are stacked on top. Because bacteria reproduce clonally, a random split places close relatives on both sides, so we withheld complete sequence types of *Escherichia coli* from training. Of fourteen reviewed predictive studies only two split by population structure, and none corrects for the fact that a lineage holdout also lowers the test prevalence, which directly affects the area under the precision-recall curve (PR-AUC). Our model achieved an average PR-AUC of 0.849 under a random split, compared with 0.803 for the gradient-boosting baseline, a margin contained within the bootstrap interval, while performance dropped to 0.634 under sequence-type holdout. After matching the holdout prevalence to the training prevalence, without retraining or removing resistant isolates, performance recovered to 0.723, so the prevalence-adjusted gap was 0.126 rather than the raw 0.215. The seven antibiotics then formed two groups, losing either 0.017 to 0.031 or 0.176 to 0.244 with none in between, and none met the applicable Clinical and Laboratory Standards Institute error limits. The contribution is therefore a protocol rather than a score: genomic resistance prediction should be evaluated with a lineage-aware split together with the prevalence correction it makes necessary, because generalisation depends on the resistance mechanism of the individual antibiotic and not on architecture alone.**

***Index Terms*—Antimicrobial resistance, *Escherichia coli*, genomic prediction, hybrid learning, lineage holdout, mixture-of-experts, multi-view learning, prevalence correction.**

I. Introduction

One of the largest threats to global health in this century is antimicrobial resistance (AMR). It prolongs the duration of illnesses, increases the cost of treatment, and raises the chances of death. The World Health Organization (WHO) has issued a warning to countries to work together. Otherwise, the advantages of the antibiotic era may be lost and we will be heading back to a time where even a common infection may be fatal [1]. Directly counting deaths from resistance is difficult. If a patient has a resistant infection and dies, the death is typically listed under the infection and not under the resistance. That is why researchers rely on statistical models built on hospital and mortality records. In 2019, an estimated 4.95 million deaths were associated with resistant bacterial infections, and 1.27 million of them were caused directly by the resistance [2]. Both figures are higher than the number of deaths from HIV/AIDS or from malaria in the same year [2]. A later analysis by the same collaboration projects that 39 million people will die of directly attributable resistance between 2025 and 2050 [3]. These are model estimates rather than direct counts.

The same trend is evident in hospital data. WHO has a programme called the Global Antimicrobial Resistance and Use Surveillance System, or GLASS. This programme enables laboratories in many countries to report antibiotic resistance of bacteria. The latest results show that about one bacterial infection in six does not respond to the antibiotic chosen for treatment. GLASS also monitors particular pairs of bacteria and antibiotics. Of these pairs, over 40% were more difficult to treat from 2018 to 2023 [4]. It is essential to make an early prediction of resistance. It helps physicians treat infections properly and helps prevent unnecessary broad-spectrum prescribing. At present, doctors rely on bacterial culture tests. These tests show how a strain responds to antibiotics, but they take 18 to 72 hours to complete. While waiting for results, a doctor may prescribe a broad-spectrum drug as a precaution, and this decision adds to the broader resistance problem. Some bacteria also cannot be cultured in a laboratory at all, and traditional approaches do not always identify the causes behind their resistance. As resistance continues to increase, tools that can predict resistance directly from sequence data are becoming more valuable.

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This study used only previously published, de-identified bacterial genome and phenotype records and therefore required no ethical approval.

A. Rationale of the Study

With the advance of whole genome sequencing, it is now possible to read the genetic sequence of a bacterial strain in detail. This has made it possible to predict resistance directly from DNA rather than waiting for a laboratory culture to grow. Many studies in machine learning have already reported high accuracy for antibiotic resistance prediction. However, high accuracy on a benchmark does not necessarily mean that a model has learned the true biology of resistance.

An important factor underlying this limitation is bacterial population structure. Bacterial populations are strongly divided into lineages, and strains within the same lineage are likely to share their genetic background and resistance patterns. This issue is well documented in the literature on bacterial association testing. Because bacteria reproduce clonally, population structure and genome-wide linkage make it difficult to distinguish a mutation that truly causes resistance from one that is merely associated with it [5]. Methods in that field now control for population structure directly within the model [6], but this correction is almost never applied in the predictive machine learning literature. As a result, a model may appear to predict resistance very well simply by identifying the lineage a strain belongs to, without ever actually capturing the genes or mutations that underlie the resistance.

In practice, this distinction matters. A lineage-based model performs well on strains similar to its training examples, but it is far less reliable on a new strain. However, a new strain is also the one a clinician knows the least about, and it is exactly there that the model's answer carries the most weight. Predictions are least reliable where they are needed most. We are therefore motivated by two goals. First, we want a robust predictive model of antibiotic resistance in *E. coli*, using genomic data as input. Second, and more importantly, we want to measure the source of that success: how much comes from true resistance mechanisms and how much comes from population structure. This will allow its predictions to be trusted more truthfully.

B. Problem Statement

Rapid and accurate prediction of antibiotic resistance remains a challenge, and this has not changed with modern sequencing and machine learning. However, there are still some issues that impact the doctor's choice. The first issue is time. A susceptibility test takes the 18 to 72 hours noted above, and longer where the isolate must first be cultured or referred, so the doctor may have to use a broad-spectrum antibiotic until that test is done. When the chosen option is incorrect, the resistance may grow even worse. The second issue is coverage. The majority of tools assess resistance to individual drugs only. Doctors, however, often have to compare a number of drugs to select a treatment. The third issue is reach. Many tools are limited to searching for resistance genes that are known. Even if a strain is resistant, a tool may not provide a useful result if the strain does not contain any of these genes.

How these models are tested is just as important as how the models are built. Bacteria are grouped into families and strains in the same family can be highly similar. Close relatives can be found in the training set and test set during a random split. This means that the model can encounter a test strain similar to one it has already encountered in training. This can help to inflate the score. A more rigorous test employs strains that lack close relatives in the training data. But this kind of test is not provided in many studies.

Explanation is another issue. Many deep learning models give a prediction but do not explain why. Due to this fact, it is difficult to tell what the model has learned. It might have learned a gene or mutation that actually causes resistance. Or it may have learned just the family of the strain. Both would seem right from the outside of the model. But the first is true biology and the second is not. What is needed is a system that is faster, easier to understand, more thoroughly tested, and able to state the basis for its answers.

This research addresses these concerns through the following research questions. **RQ1:** Whether a hybrid model that combines neural lenses with classical learners can predict antimicrobial resistance from genomic data. **RQ2:** How much each part of the model, the neural lenses, the gate, and the classical stacking layer, actually contributes. **RQ3:** Whether attention and gating weights can be used to interpret the model and identify which genes drive its predictions. **RQ4:** Whether the model's performance reflects real resistance mechanisms, population structure, or some mixture of the two. **RQ5:** Whether the model generalises to strains from lineages it has never seen.

C. Objective

We aim to construct and validate a model to predict antibiotic resistance in *E. coli* from genomic data, for seven antibiotics simultaneously. We also aim to quantify how much of its predictive capacity comes from true resistance mechanisms as opposed to population structure. Unlike a laboratory culture system, which takes a long time to produce a result, our system is designed to provide a fast prediction along with an honest account of what that prediction is based on. The work is structured in four steps to achieve this goal.

- Create a multi-antibiotic dataset controlled for leakage. We start from the publicly available Arcadia Science *E. coli* genotype and phenotype resource, which combines approximately 7,000 strains from around the world with matching resistance labels [7]. A companion study on the same dataset [8] examines the genetic basis of resistance and provides a biological framework for interpreting our results. For each genome, three views are constructed: SNP blocks clustered by gene, a gene presence/absence profile, and per-gene summary statistics. A separate collection of *E. coli* from Giessen is kept aside.
- Develop TriLens-MoE-Stack. The three views are routed through three specialised neural lenses, which are fused through a per-sample, per-antibiotic mixture-of-experts gate, and then stacked with five classical base learners through a per-drug logistic regression meta-learner. Prediction for all seven antibiotics is done simultaneously as a multi-label task. In the gene presence and absence lens we discovered an issue with the standard gated attention

and resolved it. This lens is based on a list of ones and zeros, which indicate the presence or absence of genes in a strain. With this type of input, the standard method gives the same score to every strain. So, it cannot properly separate one strain from another. We do not just say that our fix works. The benefit and the cost of it are measured by a controlled ablation.

- Evaluate generalisation on more than one kind of split. Results are shown for each of the following three splits: a random split, a phylogroup-stratified split, and a lineage holdout based on sequence type of each strain. Each sequence type is not found in more than one partition. The second split is a stratification split and not a holdout. An overall report of all three makes the difference between the designs clear. We also rebuild and retrain the pipeline from scratch on the independent Giessen collection. This collection cannot support our three-view representation used for Arcadia, since it has only a single reference feature space, so this final test checks whether our methodology transfers to a new collection, not whether an already trained model transfers.
- Measure lineage against mechanism. We combine the model's attention and gating weights with post-hoc attribution over the classical learners. The resulting marker genes are then functionally annotated and compared with a baseline given only general lineage information. Our intention was for the explanation to come from the model itself. In practice, the two sources do not fully agree, and we report that discrepancy rather than smoothing it over.

D. Scope and Challenges

The work covers seven clinically important antibiotics and treats prediction as a multi-label problem, meaning one strain can be evaluated against all seven drugs at once, just as resistance occurs in practice. Interpretability is included as a core part of the work, not an add-on. There are various ways in which generalisation is tested. Clinical deployment, however, is specifically beyond the scope of this work. The system is a research prototype, not a clinical tool. Its very major error (VME) rate, the rate at which a truly resistant strain is reported as susceptible, is above the Clinical and Laboratory Standards Institute (CLSI) recommended error rate [9] for all seven antibiotics evaluated. Those limits were written for the verification of commercial susceptibility testing systems against a purpose-built challenge panel, so we use them as an external reference point rather than as a test this work is formally subject to. It is not designed to guide patient care.

The first challenge was before the modelling stage. We required a large collection of genomes where the resistance labels were recorded in the same manner. Creating this data set was more challenging than anticipated. The raw data were in the form of variant call and PLINK [10] files, which cannot be read directly by any modelling tool. Converting them on a single machine was slow, runs took hours, and the machine sometimes crashed. The tables we built from that data had their own problems, with many duplicate and overlapping columns, and none of them were grouped by gene. We therefore switched to a published variant release instead. The dataset remained large even after this change, occasionally causing processing failures. Memory was also a limiting factor, so intermediate results were saved throughout the analysis. Once training began, the classes were highly imbalanced, with far fewer resistant cases than susceptible ones. The models were able to predict susceptibility most of the time, so accuracy looked good, but this concealed the resistant cases that were missed. We therefore gave more weight to the rare class during training and stopped judging models by accuracy alone, using precision-recall AUC instead, since it focuses on detecting resistant cases. Consequently, there was a large variation in scores among the antibiotics tested. The patterns for amoxicillin-clavulanic acid were not generalisable to new data, and this was mitigated by early stopping and regularisation. We had a well-tuned classical model too as a real competitor and not a formality.

E. Our Contributions

This report makes five contributions.

1) **A three-view model with a learned gate.** TriLens-MoE-Stack reads three genomic views, and a mixture-of-experts gate learns how much to trust each view for each strain and each drug instead of using fixed weights.
2) **A prevalence correction for lineage-aware testing.** We measure how much of the drop comes from the shift in test prevalence and separate it from the real loss of generalisation. As far as we know this is new for this task.
3) **Per-antibiotic reporting of the corrected drop.** Reported per drug instead of as one average, the drop splits into two groups with an empty space between them. A single average hides this.
4) **A fix for a problem in gated attention.** Standard gated attention breaks down on binary gene data. We correct it, and the ablation shows the fix is visible only under the lineage holdout, which supports our main argument.
5) **Honest reporting of negative results.** Attention did not explain the model, the neural part adds almost nothing on unfamiliar lineages, and no antibiotic met the clinical error limits. We report all three.

Section II reviews earlier work, Section III describes the data, Section IV describes the model, Section V gives the results, and Sections VI and VII discuss and conclude.

II. Related Work

Antibiotic resistance is the ability of bacteria to survive antibiotics that are meant to kill them or stop them multiplying. It arises in two ways: through mutations in the bacterium's own genes, or by acquiring a resistance gene from another bacterium, a process called horizontal gene transfer. Resistance has to be studied at the level of the individual strain. A strain is a variant within a species, defined by differences in its genetic code, and two strains of the same species can respond very differently to the same antibiotic. The bacterial genome is the organism's complete set of DNA. It has all the features that are utilised in this research. These are genes that it

acquires from elsewhere, point mutations in its own genes, and short sequence patterns such as k-mers and single-nucleotide polymorphisms (SNPs).

A. Methods for Combining Multiple Views and Multiple Models

The studies reviewed in Table I differ mainly in what they feed a model. Our work differs in how it combines what it is fed, so we briefly review the methods our design draws on. The mixture-of-experts idea, training several specialised sub-models and learning a gate that decides how much to trust each one, goes back to Jacobs et al. [11], and was later scaled to modern networks by Shazeer et al. [12]. The appeal for our problem is direct: different antibiotics are resisted through different mechanisms, so a fixed combination of views would dilute whichever view happens to be right for a given drug, but a gate can shift its preference from one drug to another.

Wolpert's stacked generalisation showed that the predictions of several base learners can themselves be used as features for a second-level model [13]. Training that second level on out-of-fold predictions, rather than in-sample ones, keeps it from trusting a base learner that has simply memorised the training data, which is why our meta-learner is fitted on cross-validated predictions. A consistent finding across recent benchmarks is that gradient-boosted trees remain competitive with, and often better than, deep architectures on tabular problems [14], [15]. Our architecture responds to this fact directly, by keeping a classical arm rather than replacing it. TabNet was proposed specifically to close this gap, with an attention-based architecture designed for tabular inputs [16], and we use it as a baseline in Section V for exactly this reason.

Attention-based pooling lets a model weight a variable number of items, without its parameter count growing with that number [17]. The gated variant by Ilse et al. [18] was designed for cases where a large bag of items contains only a few informative ones, which is a fair description of a pan-genome, where a strain carries thousands of genes and only a handful matter. On interpretation, SHapley Additive exPlanations (SHAP) provide a unified attribution framework with consistency guarantees [19], and we use it alongside the model's own attention weights, rather than in place of them; Section V-I reports that the two do not agree, which is itself one of our findings. Each of these methods was developed and benchmarked on data where training and test samples are drawn independently from the same distribution. That is the assumption under which a gate learns which sub-model to trust, and a meta-learner learns how to weight its inputs, but none of them was designed for the case where the test set comes from a lineage the base learners have never seen. Whether a learned gate still routes sensibly under that shift, and whether the neural and classical halves of a stack degrade at the same rate, are open questions in this literature. Section V answers both for our own model, and the answers are not the ones the architecture was designed around.

B. Evaluation Design and Population Structure

This group of studies is organised by how they test, and that is where this research makes its contribution; the relevant literature sits mostly outside the predictive machine learning field. This problem is old, and well understood, in bacterial genome-wide association studies. Bacteria reproduce by copying themselves rather than by mixing genes, so the genome does not reshuffle the way it does in sexually reproducing organisms: variants that sit far apart stay linked, and whole lineages inherit them together. Saber and Shapiro benchmarked association methods on simulated bacterial genomes and phenotypes for exactly this reason, reporting that population structure and genome-wide linkage make a causal variant hard to separate from one that is merely linked to it [5]. The response in that field has been to build the correction into the model, rather than apply it afterwards. Lees et al. fit an interpretable regression that spans the whole pangenome, so variants compete against one another inside a single model, instead of each being tested on its own, absorbing much of the signal that would otherwise be wrongly attributed to a linked variant [6].

The predictive machine learning literature has largely not adopted this correction, and the reason is partly structural. An association study asks which variant causes the phenotype, so confounding by ancestry is the central threat and has to be handled. A prediction study asks something different, namely how well the phenotype can be guessed, and here ancestry is a legitimate source of predictive signal, so nothing in the task itself forces the question. The confound only becomes visible when the model meets a lineage it has not seen, and a random split never produces that case. The evaluation-split column of Table I shows how this plays out. Fourteen of the entries predict a strain's resistance phenotype; the other two are a review [20] and a tool that classifies genes into resistance categories rather than predicting a phenotype [21]. Of those fourteen, two partition by population structure as their primary design [22], [23]. Five pair cross-validation or a holdout with an independent external collection, three use cross-validation alone, and four use a random or stratified holdout with no external set. Of the remaining twelve, two report a structure-aware analysis alongside a main result that does not use one: Moradigaravand et al. hold out the ST131 lineage in a supplementary figure [24], and Orcales et al. provide multi-locus sequence typing (MLST)-blocked cross-validation as an option [25]. Hu et al. are counted above rather than here, because they compare random against phylogeny-aware folds throughout their benchmark and not as a secondary check [23]. This is not a criticism of any individual study. But not one of the fourteen corrects the measure for the change in resistance prevalence that a lineage holdout causes, and not one separates the resulting drop by antibiotic.

Two gaps remain even after Yu et al. Their result is reported as an average across five pathogens, so it establishes that a drop exists without saying which antibiotics carry it. And accounting for structure during splitting is not the same as holding a lineage out, because a split can balance lineage composition across partitions and still leave close relatives on both sides, which measures something quite different. Section V returns to this distinction, because we built such a split ourselves before realising it could not answer the question we were asking.

C. Summary of Key Findings

Four points follow from this body of work. The first is that reported accuracy is already high, and that classical models are hard to beat. Random Forest and gradient boosting remain remarkably strong baselines, often matching or beating more complex deep architectures, and the same pattern holds on tabular data more generally [14], [15]. Convolutional and attention-based models can add value when features are well encoded, as k-mers, unitigs, or gene presence and absence, but they do not consistently outperform a well-tuned classical method. The second point, and the most important one for this research, is that evaluation is where the field is weakest. Only two of the fourteen predictive studies in Table I partition by phylogenetic structure as their primary design [22], [23], and both report a drop in accuracy once they do, so most figures in this field are figures obtained on strains related to the training set.

The third point is that most high-performing models remain hard to interpret, which limits both clinical trust and biological insight, and that external validation on a genuinely independent collection is often missing. The fourth is less obvious: the architecture literature our own design draws on has developed largely separately from the AMR application literature. Mixture-of-experts gating, stacked generalisation, and attention pooling are all mature methods, but the question of how best to combine several genomic views has received far less attention in this field than the question of which single view to use. Taken together, these four points set the direction of the present research. Rather than adding data types or scaling the model up, we work in a single well-characterised genomic setting in *E. coli* and ask how much of the predictive signal is genuine mechanism and how much is relatedness. The gaps identified here, namely honest evaluation, interpretability, and generalisation, are the ones that consistently separate a reported accuracy from a useful one.

Table I summarizes the key findings of the existing research. The evaluation-split column is the property most directly relevant to the argument above, so it is set apart from the modelling detail rather than buried among it.

III. Methodology

This study is focused on predicting which antibiotics will be resisted by a strain of *Escherichia coli* from the strain's genomic data. For this, we used a well-defined pipeline. We first assembled a large public data set that provided the genomic variation in thousands of *E. coli* strains, along with the laboratory resistance phenotypes [7], [8]. Next, we prepared the data to be used for modelling. First, records from the three source files were matched. The laboratory data was then converted to numbers. If a strain and drug combination had not been previously evaluated, then a mask was applied. A guessed value was not placed in it. From the cleaned data we generated three complementary genomic views: single-nucleotide polymorphism (SNP) blocks grouped by gene, a gene presence and absence profile and per-gene summary statistics. We then implemented a hybrid architecture, TriLens-MoE-Stack, where three specialised neural lenses compute these views, a mixture-of-experts gate combines them and their result is added to five

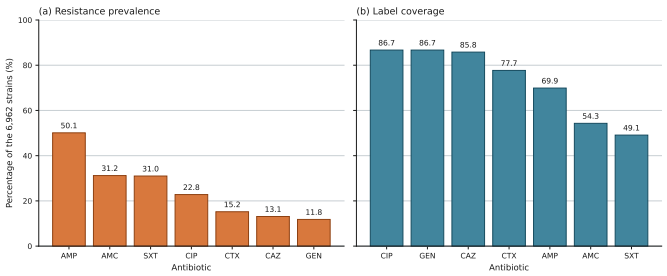


Fig. 1. Resistance prevalence and label coverage for the seven retained antibiotics. **(a)** Resistance prevalence, which ranges from 11.8% for gentamicin to 50.1% for ampicillin. **(b)** Label coverage; no antibiotic carries a label for more than 86.7% of the genomes. Both quantities are bounded above by 100%.

classical base learners via a logistic-regression meta-learner. Lastly, we considered each component using precision-recall AUC and clinical error rates. Three data splits were tested: random, phylogroup-stratified and sequence-type lineage hold-out. The method was also validated using an independent data set. In summary, the pipeline proceeds one step at a time, starting with the collection and purification of genomic data, then creating a model that can predict resistance to seven antibiotics simultaneously and is interpretable and fairly assessed.

A. Top-Level Overview of the Proposed System

The pipeline starts with an open-source genomic resource released by Arcadia Science [7], [8], which associates the genotype of each strain with its resistance phenotype that has been experimentally determined in the laboratory. Since the raw files could not be directly used in machine learning, we joined the genotype and phenotype records, standardised the column names, eliminated duplicates and incomplete records, and coded the resistance labels as integers. There are approximately 7,000 bacterial strains in the published dataset. Among them, 6,962 have complete genotype and phenotype data. These strains were used in our study. From the available labels, we chose seven antibiotics which had an acceptable ratio between resistant and susceptible samples, namely ampicillin (AMP), ciprofloxacin (CIP), amoxicillin-clavulanate (AMC), trimethoprim-sulfamethoxazole (SXT), gentamicin (GEN), cefotaxime (CTX) and ceftazidime (CAZ). Figure 1 shows how much resistance and label coverage each of the seven has.

In the cleaned dataset we have built three different genomic views which are complementary to each other, such that the model is not based on one representation. The first is a collection of single-nucleotide polymorphism (SNP) blocks, by gene, of shape $6{,}962 \times 5{,}000 \times 64$, obtained following a two-step gene filter, both stages of which are fitted on the training partition alone: a variability filter (kept only genes with at least five SNPs), which reduced 18,975 genes to 13,799, and an importance filter (ranked by gradient-boosting gain), which retained the top 5,000 genes. These combined brought 535,439 SNPs to 262,225, or about 49% of the retained variants. The second view is a gene presence/absence profile with shape $6{,}962 \times 32{,}441$, which captures resistance genes carried on

TABLE I
SUMMARY OF RELATED WORKS ON GENOMIC ANTIMICROBIAL RESISTANCE PREDICTION. THE EVALUATION SPLIT COLUMN RECORDS THE PARTITIONING SCHEME AS REPORTED BY EACH STUDY. ABBREVIATIONS: RF, RANDOM FOREST; LR, LOGISTIC REGRESSION; SVM, SUPPORT VECTOR MACHINE; CNN, CONVOLUTIONAL NEURAL NETWORK; NN, NEURAL NETWORK; WGS, WHOLE-GENOME SEQUENCING; FCGR, FREQUENCY CHAOS GAME REPRESENTATION; ARG, ANTIBIOTIC RESISTANCE GENE; MCC, MATTHEWS CORRELATION COEFFICIENT; P/A, PRESENCE AND ABSENCE; MDR, MULTI-DRUG RESISTANT; MLST, MULTI-LOCUS SEQUENCE TYPING.

Study	Data and method	Evaluation split	Result and limitation
Moradigaravand et al. (2018) [24]	Pan-genome; gradient boosting, RF, LR; 1,936 *E. coli*, 11 compounds	Random, with an ST131 lineage holdout as a supplementary analysis	Gradient boosting best; feature importance recovers *gyrA* and *parC*; the ST131 holdout scores about 0.28 below a size-matched random control
Her and Wu (2018) [26]	Pan-genome; gene-cluster scoring by genetic algorithm + classifier; 59 *E. coli*, 12 drugs	Leave-one-out cross-validation	Accessory genome carried most signal; very small collection; no lineage-aware holdout
Arango-Argoty et al. (2018) [21]	Metagenomic reads; DeepARG vs BLAST best-hit; CARD, ARDB, UNIPROT (14,933 ARGs)	Random 70/30, with external sets	Higher precision and recall than best-hit; classifies genes into resistance categories rather than predicting a strain phenotype
Ren et al. (2022) [27]	WGS, one-hot and FCGR; LR, SVM, RF, CNN; Giessen (987) and public (1,509) *E. coli*	Stratified cross-validation with an independent external set	RF best, AUC around 0.96; class imbalance; associated mutations reported for each antibiotic
Jain et al. (2025) [28]	k-mers; two-tier ML with explainable AI; *P. aeruginosa*	Stratified 75/25, with an independent 56-genome set	79% accuracy, AUROC 0.77; phylogeny used to weight features but not to split
Gao et al. (2024) [29]	k-mers; RF, SVM, XGBoost; 339 *A. baumannii*, 13 antibiotics	Ten-fold stratified shuffle-split cross-validation, with an independent 120-isolate test set	Essential agreement $>$90.9%, categorical agreement $>$95.3%, without curated genes
Do et al. (2024) [30]	Unitigs; DBGWAS, GATB, XGBoost; *P. aeruginosa* pan-genome	Ten-fold stratified cross-validation, with an independent 414-isolate collection	Novel candidate resistance genes found; AUC falls from above 0.929 in training to about 0.77 on the independent collection
Serajian et al. (2024) [31]	31-mers; MTB++ (LR, RF); *M. tuberculosis*, 13 drugs	Five-fold cross-validation	Database-free prediction; k-mer ranking computationally heavy
Yun et al. (2024) [20]	Review of machine learning for AMR in foodborne pathogens	Not applicable, review	Reports accuracies up to about 99% from the studies it surveys; small n for some drug and pathogen pairs
Lopez-Cortes et al. (2024) [32]	MALDI-TOF spectra; MSDeepAMR, transfer learning; three species	Stratified 80/20 with ten-fold cross-validation, and cross-site transfer	About 10% better than conventional ML; strong transfer-learning gains
Hu et al. (2024) [23]	Benchmarking ResFinder, Kover, PhenotypeSeeker; 78 pathogen-antibiotic pairs	**Random, phylogeny-aware and homology-aware folds**	No single tool best everywhere; scores fall markedly under phylogeny-aware folds
Wang et al. (2025) [33]	SNPs and gene P/A; XGBoost with SHAP; 296 *H. pylori*, 4 antibiotics plus MDR	Random	Consistently strong AUC; SHAP flagged 23S rRNA mutation; no external validation
Orcales et al. (2024) [25]	Genomic; LR, RF, XGBoost, NN; 1,936 *E. coli*	Cross-validation, with an MLST-blocked option	XGBoost most reliable; warns about overfitting and feature misreading
Paredes-Gutierrez et al. (2025) [34]	Mutations; XGBoost, LightGBM, ANN; 847 *M. tuberculosis*	Random 677/170, with ten-fold cross-validation inside training	Accuracy about 97% for XGBoost and 95% for LightGBM; intergenic mutations important; ANN worst
Siddiqui and Tarannum (2026)† [35]	Motifs + pan-genome; 1D CNN + XGBoost ensemble; Giessen (809) *E. coli*, 4 antibiotics	Stratified 80/20 holdout	MCC 0.926 for CIP, macro F1 0.691 for GEN; attributions concentrate on *fusA* and *parC*; assumes positional order is informative
Yu et al. (2025) [22]	Genomic; LightGBM; over 24,000 genomes, 5 pathogens	**Phylogeny-aware**	Accuracy drops when phylogenetic structure is ignored; more data does not fix it

Note: †Prior work of one of the present authors.

mobile genetic elements. The third is a collection of shape $6{,}962 \times 5{,}000 \times 4$ per-gene summaries.

The processed data were split into three parts, a training set of 4,872 strains, a validation set of 1,045 strains and a test set of 1,045 strains, about a 70/15/15 split. The lineage holdout below assigns whole groups and so lands on slightly different sizes; Table V gives the exact figures for all three. The random split randomly assigns strains, thus related strains can be found on both sides. The phylogroup-stratified split preserves an equal distribution of the eight phylogroups in each split. But each phylogroup is present in both the training and test sets. This test, then, does not measure generalisation to unseen lineages but simply the effect of the composition of the lineages. The more challenging question is answered by the sequence-type lineage hold-out split. In this split, any test

strain is not of the same sequence type as one of the training strains.

TriLens-MoE-Stack is at the center of the system. It has a feature extractor to extract features from each genomic view and pass them to a separate neural lens. The SNP blocks are fed into a convolutional network called the Motif Pattern Encoder, which identifies small and local mutation patterns in each gene. The gene presence profile goes to the Gene Feature Attention Module, which is a gated-attention network that focuses on the most informative genes. The per-gene statistics are sent to a multi-task head called the Cross-Antibiotic Context Module which provides the ability to share information across similar drugs and helps drugs with fewer labelled examples. The lenses produce a small summary vector and seven per-antibiotic scores.

A Mixture-of-Experts Gating Unit is used to concatenate the three representations into a single 192-length vector. This vector goes through a small network, followed by a softmax. It calculates the weight for each lens individually for each sample and each antibiotic. Consequently, the model is able to use various lenses for a variety of drugs. Simultaneously, the identical gene features feed five base learners, XGBoost on gene features, LightGBM, Random Forest, Logistic Regression, and XGBoost on raw features, each of which makes out-of-fold predictions using five-fold cross-validation. A logistic regression meta-learner combines the neural probability with the five classical probabilities per antibiotic, and a threshold transforms the outcome into a resistant or susceptible call. All the seven antibiotics are forecasted as a multi-label task.

We present the evaluation using four main pillars. Pillar 1 is concerned with locality. It supports the idea behind the Motif Pattern Encoder. A simple one-dimensional CNN on the entire pan-genome sequence fails the position-shuffling test. This demonstrates that positional locality does not hold across the entire genome. Convolution, however, is more effective within gene blocks. The architecture is Pillar 2. Pillar 3 is interpretability, taken from the model itself through its attention and gating weights and contrasted with post-hoc attribution. The fourth pillar is cross-dataset generalisation, which uses the same methodology on an independent Giessen *E. coli* collection. We are looking for a measure of the success of the system, not only in terms of the correct prediction of resistance, but also one that is appropriate for imbalanced data, primarily precision-recall AUC supported by clinical error rates, so that we will measure how much of the accuracy of the system represents genuine mechanism and how much of it represents population structure.

B. Data Collection

All findings in this research are based on the construction of the data, and this section explains the sources of the genomes, how raw variant files were converted to a format readable by a model, and what was lost during the conversion. We rely on the *Escherichia coli* genotype and phenotype resource published by Arcadia Science, which has over seven thousand strains, all of which were sampled all around the world, and have been lab-tested in terms of their resistance.

This resource has been published in two parts; we use both. The first release [7] was a variant calling of the collection against a pan-genome reference and the variant files were released together with the phenotype table and strain metadata. The second release [8] included gene presence and absence data for all of the genes in all of the strains, as PLINK [10] binary files. For credit and clarity, we state which view is from which release and calculate our SNP views from the first and our gene presence view from the second, not presenting the combined result as if it came from a single source. There are three reasons we chose this as a resource rather than building our own collection. The variant calls were initially quality controlled by the authoring group, effectively eliminating a significant source of variation between studies. It is part of a pan-genome and not a single genome, so accessory genes are part of the reference and in *E. coli* many resistance genes are carried on genes that the strain has obtained from other sources. And because another group has also published their own analysis of the exact same strains, we can compare results between ours and theirs later on.

The collection size varies depending on the file you are viewing: the strain metadata table contains 6,985 rows, the phenotype matrix contains 6,983, and the presence and absence release lists 7,043 samples. We made no effort to harmonize these, but retained only the strains which seem everywhere necessary, as described below.

We created a data pipeline from the raw variant files, filtering by missingness and frequency, encoding the variants as binary columns in a data matrix, then appended this to the phenotype table to get a matrix of 6,962 strains by 80,984 features before settling on this data. We changed direction due to two problems. Our filters were very crude, based on arbitrary cutoffs that had no biological significance. More seriously, the features were not structured gene-level: each variant was simply a column, and there was no connection to the gene it was a part of, and the entire concept of our proposed model is that mutations within the same gene should be read jointly. For this reason, we have decided to adopt the curated release above and rely on it from here on.

C. Data Cleaning

This primarily referred to record matching between files and encoding text phenotypes as numbers. The genomes themselves needed no cleaning, since the variant calling had already been done and checked.

1) Strain Matching Across Files: Three files were required to match for a strain to be used. The variant file is indexed by sequencing accession. These are identified in the phenotype file by genome name. They are mapped via a separate mapping file. An accession was kept if it was found in the variant file, the mapping file provided a genome name, and that genome name appeared in the phenotype table. Following this, 6,962 strains remained and these strains were used for all experiments in this research. We also incorporated the strain metadata table, 6,821 of which (98.0%) matched. Since the metadata is only utilized to work out lineage, the strains without it are not lost; they simply fall outside of a lineage group.

TABLE II
THE SEVEN ANTIBIOTIC LABELS AFTER CLEANING. COVERAGE IS THE SHARE OF THE 6,962 STRAINS TESTED AGAINST THAT DRUG; PREVALENCE IS THE SHARE OF THOSE THAT WERE RESISTANT.

Drug	Res.	Susc.	Not tested	Tested	Cov.	Prev.
AMP	2,439	2,429	2,094	4,868	69.9%	**0.501**
CIP	1,377	4,656	929	6,033	86.7%	0.228
AMC	1,180	2,598	3,184	3,778	54.3%	0.312
SXT	1,060	2,358	3,544	3,418	**49.1%**	0.310
GEN	710	5,327	925	6,037	86.7%	**0.118**
CTX	823	4,586	1,553	5,409	77.7%	0.152
CAZ	780	5,193	989	5,973	85.8%	0.131

Note: Trimethoprim-sulfamethoxazole has the lowest coverage of the seven drugs at 49.1%, and prevalence ranges from 0.501 for ampicillin down to 0.118 for gentamicin, the two spreads the next paragraph discusses.

2) Phenotype-to-Label Encoding: When a test is performed, the result is recorded in the phenotype file as resistant, intermediate or susceptible and a blank is indicated where no test was performed. This was transformed to a binary label with 1 for resistant and 0 for susceptible. In order for our labels to be comparable with the ones produced by the group that published the dataset, we encoded intermediate results as resistant, following the same convention as they initially used. There is also a clinical argument for it: an intermediate result implies that the normal dose will be insufficient, hence it is safer to call it resistant. One possible consequence of these early steps is worth remarking on: this step leaves no intermediate category, which is the reason that categorical agreement collapses to plain accuracy in Section V.

3) Treatment of Untested Strains: Not all strains were tested against all drugs, and the number of drugs tested varies a great deal from strain to strain. So, instead of discarding incomplete lines or inserting guesses, we retained a boolean mask with shape $6{,}962 \times 7$ that indicates which entries are real. Training loss and all evaluation metrics are calculated only for the observed entries. Nothing is imputed anywhere in this research. Table II displays what remained for us.

There are two items in this table that represent a lot of what happens later. Overall, almost a twofold variation exists in coverage. Ciprofloxacin and gentamicin were tested for almost 87% of strains, but trimethoprim with sulfamethoxazole for only 49%, and fewer strains tested means fewer samples behind a score, which equals a wider confidence interval. It is the chief cause of some of our intervals in Section V being so much wider than others.

Prevalence ranges more widely: from 0.501 for ampicillin, down to 0.118 for gentamicin. This is important because the no-skill value of precision recall AUC is the prevalence: a score of 0.70 on gentamicin, where only 12% of strains resist, is a much larger gain over chance than the same score would be on ampicillin. Four other antibiotics were present in the phenotype file but dropped. They either were too small to be tested or were so biased that a model would score well by making too many predictions to one class. The seven we kept cover five classes of drugs.

TABLE III
VARIANT FILTERING DURING CONSTRUCTION OF THE SNP MATRIX.

Stage	Variants
Streamed from the variant file	2,451,184
Removed: insertion or deletion	43,799
Removed: minor allele frequency below 1%	1,871,946
Kept	**535,439**

Note: The 535,439 kept variants are 21.8% of the 2,451,184 streamed from the variant file, and this is the set that feeds every later step.

D. Data Transformation

1) SNP Matrix Construction: The variant calls and the gene presence and absence profiles are released in two different deposits. As a result, the first and second views below are generated from different source files, instead of one source file. The size of the variant file is approximately 1.1 gigabytes in compressed VCF. We streamed it just once, and only retained a variant if two requirements were met. This had to be a single-nucleotide change, so insertions and deletions were omitted. In addition, it had to have a minor allele frequency of at least one percent across the collection. What this left is recorded in Table III.

The team that constructed this data set reported approximately 2.4 million variants, and we read 2,451,184, so we knew we were reading it correctly before building anything on top of it. When the frequency filter was applied alone, 76% of the variants were removed, with the majority of them present in just two or three strains. If a pattern is found in so few strains, then it cannot be expected that it will hold true for a new strain. So removing these variants was very cheap for the model.

Genotypes are represented by small integers: 0 is the reference allele, 1, 2 or 3 are the first, second or third alternate. Overall, 83.3% of the call types in the data are reference calls and 16.3% are first alternate; 2.5% are second or third alternate, among non-reference calls. We pass these through as integers and not in separate columns. If the alleles are treated as integers, then allele 3 is further away from the reference than allele 1. There is no biological basis to this. Second and third alternate alleles are, however, only 2.5% of non-reference calls, about one in forty. One-hot encoding them as separate columns was therefore not worth the added complexity.

For genotypes missing, they were set to the reference allele in line with the original authors so that our features match theirs. This is not a neutral choice: a genotype that failed to call is assumed to match the reference and coverage is typically lower on plasmids and other mobile elements that are less able to assemble from the short reads, which is where acquired resistance genes typically reside. This forms part of our explanation as to why gentamicin is the hardest of the seven drugs to model. The result is a matrix of $6{,}962 \times 535{,}439$ stored as 8-bit integers, 3.73 gigabytes, which took about two and a half hours to build on a single consumer GPU.

2) The Gene Presence and Absence Matrix: Release 2 provides the presence of genes in 32,441 pan-genome loci in PLINK binary format. It took one decision that was not necessarily obvious to make these into a usable matrix. The

numbers stored are 0 and 2. Nothing in the release specifies which one means the gene is present and the two readings have opposing biology, so we tried both against something we could test. If non-zero is considered present, the median gene prevalence across loci is 0.966 and the average strain then carries about 28,700 of the 32,441 genes. If zero is coded as present, the median gene prevalence is 0.034 and the average strain carries 3,709 genes (range 2,141 to 8,333). The two statistics are reported side by side and not derived from one another: the median is taken over loci and the count over strains, and because the distribution of gene prevalence is strongly skewed the count is not the median multiplied by 32,441. An *E. coli* genome contains about 4-5 thousand genes, limiting the reading to the second one, which is the one we use.

We register this check because the incorrect option need not have thrown any error. The model would have learned as usual with its gene matrix inverted and since this view has a lot of weight in our final model, everything further down it would have been incorrect in a non-obvious way. The per-strain gene count is a sanity check and is now done automatically on every new matrix build. There are 6,962 strains of which 6,949 are found in the PLINK sample list. The other 13 (0.19%) do not have a record of gene presence. In the default policy of the pipeline these rows are not dropped or masked, instead remaining as an all-zero gene-presence vector, meaning that Lens B interprets them as a strain with non-detected genes instead of excising them from the dataset. This conflates a missing record with a genuinely empty accessory genome, which is not the same thing, and the correct handling is to extend the label mask of Section III-C to cover the gene-presence view for these rows. At 0.19% of the collection, and with the 13 strains distributed across partitions in proportion, the effect on any figure reported here is below the resolution of the intervals in Section V-B, so we record the defect rather than reporting a corrected set of results.

3) Per-Gene Statistics: The third view summarizes each gene with four values calculated from its SNPs: mean value of the alleles, variance of the alleles, number of positions that do not match the reference, and maximum value of the alleles. It is a very broad perspective, and it cannot tell where a mutation is occurring in a gene. It does offer stability and compactness though. It also provides a brief and reliable gene-level view of variation which complements the other two views.

Each of these four statistics is on a different scale. Therefore, each one is adjusted in order to have zero mean and unit variance before it gets to the third lens. Only the training partition is used to calculate the means and standard deviations. They are then applied unchanged to the validation and test partitions. This means that no test genome impacts the scaling.

4) Lineage Encoding: Bacterial populations are structured in lineages and strains within a lineage are likely to share their genetic background and resistance pattern. In order to determine if our model was using lineage or mechanism we first had to determine the lineage of each strain, which the data set did not explicitly give us, but we worked it out ourselves. The phylogroup, in the sense of the Clermont typing scheme [36], was not provided directly. We have developed it in two steps. An explicit phylogroup assignment, if any, is taken from the typing field in the strain metadata. In the absence of such information, the multi-locus sequence type and mapping to a phylogroup were done with the help of a look-up table of 57 common multi-locus sequence types. There is no record of the origin of this particular table in the original notebooks used for data preparation. No citation or generation script is recorded with it and it is reported here as an undocumented lookup, not as a part of a published scheme. It is therefore weaker evidence than a direct assignment would be: the established route is to call the phylogroup from the genome itself with an *in silico* implementation of the Clermont quadruplex, such as ClermonTyping [37], and a study that depends on the phylogroup as more than a coarse covariate should do so. If a strain is found that cannot be fixed by either route it is marked Unknown. Table IV gives the result.

TABLE IV
PHYLOGROUP COMPOSITION OF THE 6,962 STRAINS.

Phylogroup	Strains	Share
B2	3,542	**50.9%**
Unknown	1,697	**24.4%**
D	723	10.4%
A	317	4.6%
F	307	4.4%
B1	258	3.7%
C	114	1.6%
E	**4**	0.06%

Note: B2 alone is 50.9% of the collection, 24.4% of strains have no assigned phylogroup, and phylogroup E has only 4 members, too few for any per-group estimate; these are the three limits the next paragraph discusses.

There are, however, three features of this table which restrict what can be read from it; we should like to mention them here rather than have them pointed out later. B2 is slightly more than half the collection, so the analysis of splitting the collection by phylogroup is a majority analysis of B2 vs. the remainder of the collection. There is little use for a per-group estimate of the diversity of the Unknown group because almost a quarter of strains could not be assigned to any group, and phylogroup E has a small number of members (four), which is too small to make any estimate meaningful. For this reason we consider the phylogroup as a broad category and not anything else. It is encoded as an eight-way one-hot vector and used as a covariate and as a grouping variable. We do not consider it as a fine description of ancestry.

E. Data Integration

We had two checks before training. The three feature matrices were of different files. They were also produced in various stages. We therefore made sure that they had the same strains in the same order. We also verified that they were using the same names of genes. The first confirmed that the ordering of all 6,962 strain identifiers is identical in all views; otherwise, one genome's SNPs would be paired with another genome's genes.

The latter one discovered a real challenge. None of the genes were found in common when the gene identifiers in the

SNP views were compared to the presence and absence view, because the PLINK reader had created its own placeholder names for the genes and not read the actual gene identifiers from the annotation file. Once the reader was corrected, all 5,000 genes chosen for the SNP views were found in the 32,441-gene catalogue, a 100% overlap. Nothing would have failed visibly had this check not been run. The model would have received views labelled with the wrong gene names, and the gene-level reading would have been comparing labels with no meaning against one another. Both the checks are now programmed as a fixed step prior to any training run.

F. Data Reduction

There are 535,439 features for 6,962 strains (some 77 candidate features for each sample). So, it was not an optimisation to reduce the feature space. It was necessary for modelling. The features were reduced at the gene level rather than at the individual-variant level as the model operates at the gene level. Lastly, only the genes that survived the two filters were retained. If a gene has fewer variable positions, we cannot support a within-gene pattern, which is what our first view is designed to find; thus the variability filter retains a gene if it has five or more SNPs within the training partition of the split being fitted. After this step, 13,799 of the 18,975 genes in the matrix remain on the random split. These genes are then sorted by importance in the gradient boosting model [38] using the importance filter. That model is trained separately for each of the seven antibiotics. First, we filtered each set of SNPs with a chi-squared pre-filter and kept the top 50,000 per drug, and trained a single XGBoost model for each antibiotic on its respective chi-squared filtered subset, calculating each gene's cumulative gain for each model. The final importance score for each gene is the sum of the per-antibiotic importance over all seven models, and the 5,000 most important (highest summed) genes are retained.

Both of these filters were fitted solely on the training partition, and refitted from scratch for each of the three splits. This is quite easy to forget, and quite hard to do well: had the test labels been visible during ranking, they would have changed what features exist in the first place, and each score in Section V would then have been overoptimistic by an unknown amount.

One earlier stage does not meet that standard, and we state it rather than leave it to be inferred. The variant-level filters of Section III-D, namely the removal of insertions and deletions and of variants with a minor allele frequency below one percent, were computed once over the whole collection, before any split existed. They use no phenotype label, so they cannot transmit outcome information, but they do let the allele frequencies of the test strains influence which columns are available to the model. We regard this as a weaker form of leakage than a label-aware filter would be, and we did not repeat it per split because doing so would give the three splits different feature spaces and make their scores incomparable, which is the comparison the paper is built on. Figure 2 shows how much the gene feature space is reduced by the two gene-level filters used above.

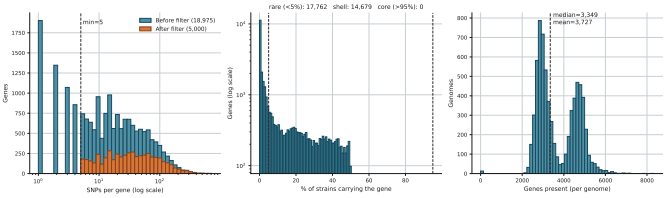


Fig. 2. The gene feature space before and after filtering. **(a)** SNPs per gene across the 18,975 genes of the SNP view, before and after the variability and importance filters cut them to 5,000. **(b)** and **(c)** the separate 32,441-gene presence and absence catalogue, which is mostly rare genes — 17,762 are carried by under 5% of strains and none by more than 95%, and a typical genome holds a median of 3,349 of them against a mean of 3,709.

After filtering, 5,000 genes remain. These genes contain 262,225 SNPs. This corresponds to approximately 49.0% of the 535,439 variants retained after the variant-level filters, and to about 10.7% of the 2,451,184 originally streamed. These genes differ widely in size: the smallest carries 5 SNPs, the median 32, the mean 52, and the largest 717. That distribution fed one last decision. Our first view reads each gene as a fixed length block, which we set to 64 positions. Approximately 28% of the selected genes are longer than 64 SNPs and are truncated, the remaining genes are padded with zeros and masked with zeros, so the padding does not contribute. A gene block has an average of 35 positions out of 64 that are occupied by real positions. Most genes would waste memory if the block were too long, while a shorter block would cause the median gene to be cut.

G. The Three Evaluation Splits

The splits determine what a score measures, because of which we give a detailed account of how we split the data. Approximately 15% of each split is for testing, another 15% of each split is for validation, and approximately 70 percent is for training. The test set is never used in the model selection or tuning; it is not touched until the final evaluation, and every split has been performed with the same fixed random seed (42), which means that any number reported in this research can be exactly re-generated by running the same code. We constructed three splits and they address three questions.

The random split assigns strains via shuffling, stratified on the ampicillin label to achieve class balance across the three sets. This results in a training set of 4,872 strains, a validation set of 1,045 and a test set of 1,045. It is the normal setting, and the one that is used in the majority of published work; the related strains can fall on both sides, and do. The partition sizes remain the same but the partition balancing takes into account the eight-way phylogroup assignment, each lineage group being represented in each of the three sets in the same proportion as it would be in the collection as a whole. This does not control class balance. The proportional guarantee is meaningful for the six larger groups; for phylogroup E, which has four members in the entire collection, a 70/15/15 allocation places at most one strain in the test partition, so that group is present in the split by construction but carries no evidential weight.

We just want to clarify what this second split is, as it can be very easily inflated. It is a stratified split, as opposed to a holdout. By design, every phylogroup appears on both sides so it is not testing how the model performs on a lineage it has never seen before. What it tests is the extent to which the result is influenced by the choice of stratification variable. This is a weaker question than the one we wished to answer.

The train/test split answering that more stringent question is the sequence-type lineage holdout, and is the one upon which the central result of this research is built, so we will describe its construction in full. Strains each have a multi-locus sequence type, a generic label in bacterial genomics that clusters strains based on shared descent in the recent past and is much more specific than phylogroup. We did not call sequence types ourselves. They were taken from the sequence-type field of the strain metadata distributed with the primary collection [7], which reports the seven-locus Achtman scheme for *E. coli*. Because the metadata table and the variant matrix are ordered differently, the join was made on the sequencing accession carried by both, and not on the position or row of a strain in either table; that distinction matters, because joining by row silently mislabels the lineage of every strain and would invalidate the split this paper is built on. The join recovered a sequence type for the large majority of the collection. Two consequences follow from taking the types as given. The scheme resolves clones but not clonal complexes, so two sequence types that are single-locus variants of one another, as several members of the ST131 complex are, can still be separated across partitions, and the holdout is therefore a lower bound on lineage novelty rather than a guarantee of it. And any error in the upstream typing propagates into our split without our being able to detect it.

Two rules were set prior to examining any result. In the absence of a sequence type, any strain was assigned to training as an unlabelled group cannot be held out safely. If the number of sequences in a collection type was more than five percent, then it was also put in training. The other rule is equally important. Type 73 represents approximately an eighth of the entire dataset. If that clone has a relatively large number of different strains found in the test set, the test will mostly be measuring this particular clone. This does not measure how well the model will work on novel lineages. The rest of the sequence types were then allocated whole to training, validation or test by a seeded permutation walk which successively adds groups until the desired proportion of genomes in each partition is reached. Figure 3 shows the size distribution these rules had to work around.

We kept whole groups in the same partition. Owing to this, there are no ideal percentages of the split. A chance division would appear less lopsided. This effect is reflected in our data split as well. The holdout uses a training set of 4,836 strains, a validation set of 1,081 and a test set of 1,045. No sequence type was observed to be present in more than one partition, as opposed to assuming this. The same point of method is used for all three splits. The 5,000 gene selection mentioned earlier under Data Reduction is re-fitted on each split's training partition instead of being computed only once and used for each split. The definition of the pipeline used an earlier version that leaked information between splits, which was fixed prior to any results in this research. Refitting per split indicates that each split generates its own ranking of informative genes. Notably, the holdout's gene set overlaps the random split's by less than half, indicating that the resistance signal is not located in a few genes but in many genes that are partly interchangeable.

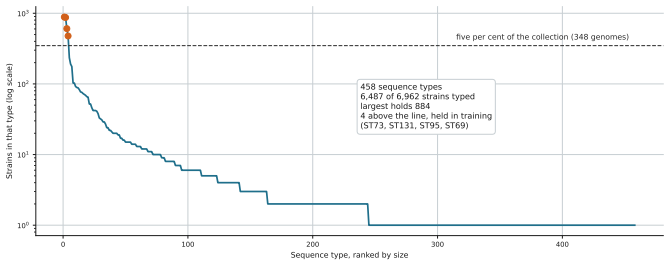


Fig. 3. The 458 sequence types, ranked by size. A few are very large and most are very small. The largest holds 884 strains, nearly one genome in eight, and four types sit above five percent of the typed collection. Those four are placed in training, so that no test set can be dominated by a single clone.

TABLE V
THE THREE EVALUATION SPLITS AND THE PARTITION SIZES THEY PRODUCE. ONLY THE THIRD ASSIGNS WHOLE GROUPS, WHICH IS WHY ITS SIZES ARE NOT ROUND.

Split	Balanced on	Train	Val	Test
Random	Ampicillin label	4,872	1,045	1,045
Phylogroup-stratified	Eight-way phylogroup	4,872	1,045	1,045
Sequence-type holdout	Whole sequence types	4,836	1,081	1,045

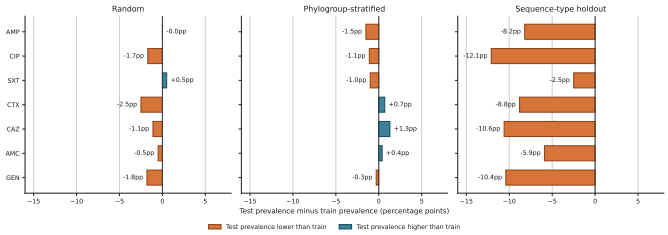


Fig. 4. Resistance prevalence in the test partition minus the training partition, in percentage points, per antibiotic and per split. The random and phylogroup-stratified splits stay within 2.5 points and move in both directions. The sequence-type holdout moves downwards for every antibiotic, by 2.5 to 12.1 points, and since the no-skill floor of precision recall AUC is the prevalence itself, part of the drop reported later follows from this shift rather than from the model.

For each split we also record the number of labelled strains and the resistance prevalence in each partition, per antibiotic. The lower bound of precision recall AUC will be determined by prevalence, meaning that a lower prevalence will result in a lower score even if the model is equally good. The effects are greatest for the holdout, as forcing the largest clones into training causes a shift in prevalence downwards for all of the antibiotics. Reporting the prevalence with the score enables a reader to distinguish a drop because of a tougher test set from a drop because of a lower floor. Figure 4 shows how large that shift is for each split.

Now combining the three genomic views with the encoding of the phylogroup, the labels and their mask provides the six arrays that are actually passed to the model, which are

TABLE VI
ARRAYS PRODUCED BY THE PREPROCESSING PIPELINE: FOUR FEATURE REPRESENTATIONS, FOLLOWED BY THE LABEL MATRIX AND ITS MASK.

Representation	Shape	What it captures
SNP blocks by gene	$6{,}962 \times 5{,}000 \times 64$	mutation patterns inside genes
Gene presence/absence	$6{,}962 \times 32{,}441$	genes acquired from elsewhere
Per-gene statistics	$6{,}962 \times 5{,}000 \times 4$	coarse gene-level variation
Phylogroup one-hot	$6{,}962 \times 8$	broad lineage
Labels	$6{,}962 \times 7$	resistant or susceptible
Label mask	$6{,}962 \times 7$	which labels were measured

TABLE VII
COMPOSITION OF THE EXTERNAL GIESSEN DATASET.

Antibiotic	Resistant	Susceptible	Prevalence
CIP	366	443	0.452
CTX	358	451	0.443
CAZ	276	533	0.341
GEN	188	621	0.232

contained in Table VI.

Two characteristics of this data set set the stage for everything that follows. It is broad and short: there are still far more features than strains even after dimensionality reduction, so the design favors gradient boosting over a large network. And it is patchy: antibiotic coverage ranges from 49 to 87% according to drug while prevalence ranges from 12 to 50%, meaning no averaged score across the seven antibiotics serves it well; hence each result in Section V is reported per drug, as well as on average.

H. The External Giessen Collection

A single collection does not provide an answer to the question of whether or not an approach travels. We use a second, independent set of *E. coli* genomes that were collected at the University of Giessen and published by Ren et al. [27]. Isolates were from human and animal clinical specimens and susceptibility was done by automated testing and interpreted according to European Committee on Antimicrobial Susceptibility Testing (EUCAST) breakpoints [39]. Ren et al. report 987 Giessen isolates in total; the subset we use is the 809 that carry a complete result for all four of the antibiotics shared with our panel, which is the same 809 used by Siddiqui and Tarannum [35], prior work of one of the present authors. Those 809 strains are described by 60,936 SNP positions each. Its composition is shown in Table VII.

There are two disparities with the Arcadia data that influence its use. The first one is the reference. In Giessen variants are listed in order of a single reference genome (columns are ordered along the chromosome) while Arcadia uses a pan-genome (the ordering of the columns has no meaning). This difference is not an inconvenience: it makes the locality test possible because we have one set of data for which locality should matter, and one set for which it should not. The second is the two feature spaces have no overlap. The model trained on Arcadia cannot be used at Giessen, since there is no shared feature vector, so anything we do across the two collections must be a comparison of methods.

Also, there is a difference in the label names. The Giessen phenotypes are interpreted against the EUCAST breakpoints, and the Arcadia phenotypes are interpreted according to a different standard. Breakpoint differences between the two also exist for the same drug, and this could contribute to performance differences between collections, and not only to the model. We are unable to separate the two effects with the data available and designate it as a limitation for any cross-collection comparison. This dataset consists of 809 strains with results for all four types of drugs, unlike Arcadia where some of the labels are missing, so masking is not necessary. The numbers were randomly split by ciprofloxacin into 647 for training, 81 for validation and 81 for testing. The test partition is small and that is why the Giessen confidence intervals are wider than those for Arcadia.

IV. THE TRILENS-MOE-STACK ARCHITECTURE

The proposed model TriLens-MoE-Stack is a hybrid architecture in which a stacking meta-learner is used to join classical machine learning models with neural lenses. The name illustrates the design: *TriLens* comes from the three genome perspectives the model uses. *MoE* stands for Mixture of Experts, where a gate determines how much each view should contribute. *Stack* refers to how the outputs of multiple classical models are combined to produce the final prediction. The concept is that a bacterial genome can be defined in more than a single manner and no single definition is complete by itself. A mutation-based stance will lose a resistance gene that came into the plasmid, since that gene is not a mutation, it is only there. A point mutation in a target enzyme is not detectable by the gene presence view. This is because the gene is present in both resistant and susceptible strains. The only difference is a single nucleotide change. Instead of selecting between these views, our design incorporates all three and defers to the model to determine the importance of each view for each antibiotic.

This system has three components and the entire architecture is presented in Fig. 5. To begin with, the three genomic views are processed by a neural feature extractor that uses three specialised lenses and combines them with a mixture-of-experts gate. We then use the gene features to train five classical base learners. They make leak-free out-of-fold predictions. Third, a logistic-regression meta-learner combines the neural and the classical predictions into a single per-antibiotic prediction.

In the model, each component generates a raw score z, whose value may be any real number, and the score is converted into a probability in the range 0 to 1 with the help of the logistic sigmoid function,

$$\sigma(z) = \frac{1}{1 + \exp(-z)}, \tag{1}$$

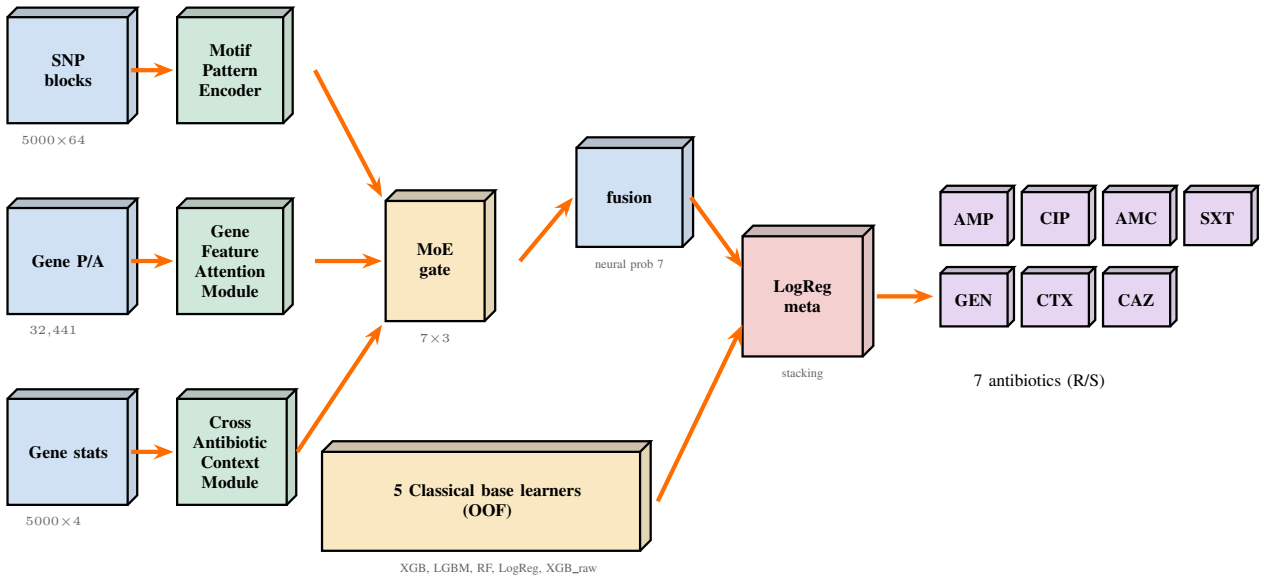


Fig. 5. Overall dataflow of TriLens-MoE-Stack: three genomic views feed three lenses, which the mixture-of-experts gate fuses before stacking with five classical base learners.

A. Rationale for the Hybrid Design

Combining neural and classical models is one of the most crucial design decisions. We use both instead of one deep neural network. An apparent solution in the contemporary world would be an end-to-end trained network. We have experimented with that, and it was not very good, but the reason is in the form of the data, and not in anything wrong with the network. Deep learning works best in situations where the number of training examples is significantly larger than the number of input features; our scenario is the opposite with only 6,962 labelled strains and hundreds of thousands of candidate genomic features. The training data can also be memorised in a model that can utilise all this information. Even with this number of samples, dropout cannot completely avoid that issue.

Gradient boosted trees are much more suited for this type of data. For a feature containing no signal, the split predictor will just never consider that feature for a split, and it requires far fewer examples to train a stable model. The classical arm, then, is not a fallback or a leftover baseline, it is a load-bearing part of the final design. Meanwhile, classical models regard each feature as an independent column: they cannot encode the idea that various mutations near each other within a single gene function as one pattern, nor that resistance to one cephalosporin says something about another. Those are precisely the types of tasks a neural network excels at, meaning that the design pairs both, with each arm addressing the other's weakness. Section V provides the proof: on the random split the neural arm alone hits 0.781 while the combination 0.849, and under the lineage holdout the classical learners alone score 0.628 while the full model gets 0.634, but the neural arm yields only 0.472.

B. Classical Base Learners

The classical arm comprised five base learners: the XGBoost on the gene-aggregated features (XGB_gene), LightGBM, Random Forest, Logistic Regression and a second XGBoost on a chi-squared-prefiltered raw feature matrix (XGB_raw). These five are trained in an independent five fold stacking pipeline, and the five out of fold predictions are provided to the meta-learner of Section IV-E. Each of the five on their own is presented as a baseline for the proposed model in Section V.

Five models might appear to be more than is needed so it is worth explaining why. Only when models of an ensemble commit a variety of errors does the ensemble work. When two models commit similar errors, their combination does not give much advantage. It averages just two similar false forecasts. We thus selected five learners who fail in various manners. Both XGBoost and LightGBM are gradient boosting methods, but they expand their trees in different ways, thus overfitting to different aspects of the data. Random Forests use multiple trees, which are averaged and produced to provide a low-variance but less sharp model. Logistic Regression is also linear and is unable to model interactions in any way, but it is highly stable. The second XGBoost does not require gene-level summaries but instead works with raw single-variant features, thus able to capture signal that is at a single position and would be lost through aggregation. The five are differentiated between in the three ways that are relevant here: whether a model is able to pick up combinations of features or only weighs each feature separately, the extent to which its predictions change when the training data is perturbed slightly, and whether the model reads the genome one gene at a time or one variant at a time.

Five-fold cross-validation is used to train each base learner, thus each strain is predicted using a model that was never trained on it. This is more important than it may sound as a model trained on the full dataset would give overly high scores

when asked to make predictions and the meta-learner would end up putting too much trust in those inflated predictions. This leak is avoided by out-of-fold prediction, which is why the weights of the stacking layer can be believed.

1) XGBoost: The gradient boosting idea [40] specialised for speed and scale is XGBoost [38] that builds a prediction from multiple shallow decision trees one by one, each trained from the residual errors of all preceding trees. Formally, the prediction score for a sample x is the sum of the outputs of T trees:

$$\hat{y}(x) = \sum_{t=1}^{T} \eta \, f_t(x), \tag{2}$$

with each f_t being one tree and η being a small learning rate, or shrinkage, which reduces the contribution of each tree in the model to ensure no one tree is dominant and the model is enhanced in small, steady steps. Training will reduce a loss that is penalised:

$$\mathcal{L} = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{t=1}^{T} \Omega(f_t), \tag{3}$$

and $\ell(\cdot)$ is the binary cross-entropy loss on the $\sigma(\hat{y})$ and $\Omega(f_t)$ is the penalty on trees with a large number of leaves or an excessive number of extreme leaf values, thus any extra complexity must decrease the error by an amount that justifies its cost.

There are two reasons for XGBoost's inclusion in this list as far as this data is concerned. If a feature has no information, it is not split, thus a tree can represent a very wide matrix, and if there is a resistance that requires two mutations together, the tree can split on one and then the other, which no linear model can do. If the number of susceptible strains is much higher than the number of resistant strains for some drugs, scale the pos weight according to the ratio of the susceptible to the resistant samples. A summary of the genome is represented using XGBoost on the gene-aggregated features, while a fine-grained view is represented by XGBoost on a chi-squared-prefiltered raw matrix. We choose the number of trees using early stopping [41].

2) LightGBM: LightGBM [42] applies the same boosting concept with a couple important shortcuts that are particularly relevant when the data is this wide: histogram binning, which only uses the bin edges when searching for a split point, and leaf-wise growth, where after each boosting round it expands whichever single leaf most reduces the error rather than growing level by level. The outcome is a deeper tree in the areas where the data is sparse.

Training does not begin from scratch. The initial score μ is the mean resistance rate of the training data, and each tree after that is multiplied by a learning rate η before it is added. Due to this, an overall score for a strain will be

$$F(x) = \mu + \sum_{t=1}^{T} \eta \, f_t(x), \tag{4}$$

Here $\eta = 0.05$ and the raw score is transformed into a probability using (1).

The obvious objection is that adding a second gradient boosting model to XGBoost contributes nothing. The two behave differently, and both are worth keeping. Finally, LightGBM constructs trees in a leaf-wise manner. This enables it to produce extremely imbalanced trees. Consequently, it is able to pick small and specific patterns that the more balanced trees in XGBoost do not pick up. As a result, the two make different errors, which is precisely what a stacking ensemble can exploit.

3) Logistic Regression: The simplest model of the ensemble and the easiest to explain is a Logistic Regression [43]. It assigns a weight value to each feature and then multiplies the value of each feature by the weight. It then sums all the numbers and feeds it into a sigmoid to get a probability. There are no trees, feature interactions or hidden layers in the process. When a mutation is linked to resistance the weight is positive, when to susceptibility the weight is negative, and the magnitude of the weight is an indicator of the strength of the effect. In the model, the features are linearized to give a score,

$$z = w^{\top} x + b, \tag{5}$$

where w is the weight of the features and b is a bias that changes the aggregate baseline and the score is transformed to a probability using (1). Training is used to minimise penalised binary cross-entropy loss,

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right] + \lambda \, R(w), \tag{6}$$

with the first term preferring to penalise certain confident errors more than uncertain ones, and the second term, $R(w)$, ensures that the weights are kept small in order to prevent any single feature dominating.

The thing is its simplicity. Since it cannot represent interactions, it can not invent interactions, and consequently overfits much less than the tree ensembles: when the boosting models become over-confident on a given strain the linear model remains closer to the average and the meta-learner can learn on it.

4) Random Forest: Random Forest [44] operates in the reverse direction to boosting: a large number of trees are constructed and averaged, not in series. Two sources of randomness ensure that they are not duplicates of each other, a bootstrap sample of the strains [45] per tree and random selection of features to be explored at each split, making their respective errors point in different directions and most of them cancel. Given a sample x, the predicted probability of resistance is the average of the per-tree probability estimates:

$$\hat{p}(x) = \frac{1}{T} \sum_{t=1}^{T} p_t(x), \tag{7}$$

where $p_t(x)$ is the proportion of the leaf in the tree t that x visits that is of class one. Graded probabilities are meta-features that would not be provided by a hard majority vote, and are required by the stacking layer. Random Forest adds stability instead of optimum accuracy. The predictions are relatively robust to slight changes in the training data, which

is a property that boosting models in general do not have, and thus the ensemble will certainly possess an element that is hard to push to one extreme.

C. Neural Lenses

The neural feature extractor has three lenses that correspond to the three genomic views. Each reads its respective view, creates a short representation in a 64-length summary vector z, and outputs seven scores, for each antibiotic. All three struggle with the same challenge: each strain is characterized by thousands of genes, but most of those genes are not relevant for any one antibiotic and which ones are relevant changes with drug. This calls for a mechanism that can focus on a handful of pertinent genes without any prior knowledge as to which. This mechanism is the gated attention of Ilse et al. [18], which is an extension of the attention mechanism [17] for the scenario where only a few items in a large bag are informative. Every element h is given a relevance score,

$$e = w^{\top}\big(\tanh(Vh) \odot \sigma(Uh)\big), \tag{8}$$

and the U, V and w are learned in training and $\odot$ represents element-wise multiplication. The expression has two branches for a reason. The $\tanh$ branch puts forward the degree to which this element appears to be valuable and results in a value between -1 and 1. The σ branch is a gate which generates a value in the range 0 to 1 that determines the portion of that proposal to permit through. An element may thus be suppressed even though it may appear on the surface to be informative which cannot be done by ordinary attention. The scores are normalised with a softmax giving them weights which sum up to one and the pooled summary is the weighted sum of the elements. In short, each lens learns to summarise a very large input by selecting the few elements that carry the signal.

1) Motif Pattern Encoder: The Motif Pattern Encoder reads SNP blocks, whose shape is (G, L) where $G = 5{,}000$ genes and $L = 64$ positions per gene. It is used to identify short local motifs, or patterns, of mutations within a gene. The reasoning is that one mutation is generally silent, but then two or three mutations in the same gene, close together in the DNA, are more likely to break or change the protein that the gene codes for. The natural device to this is a convolutional network, since a convolution is performed by sliding a small window over a sequence and finding a pattern anywhere in the sequence. Each gene has two convolutional layers stacked:

$$h_g = \mathrm{ReLU}\big(\mathrm{Conv1d}_{16\to 32}\big(\mathrm{ReLU}(\mathrm{Conv1d}_{1\to 16}(x_g))\big)\big)\,, \tag{9}$$

with windows of width three, with dropout [46] between layers. The initial layer is trained on 16 simple patterns and the second layer is trained on 32 more complex patterns based on the simple ones, thus, the model is constructed to be based on simple motifs and then compound motifs. A masked max-pooling step then prunes each gene to a fixed length vector by retaining, in each learned pattern, the strongest response it could find anywhere in that gene. The mask makes sure that padding positions in shorter genes are not considered as real data. The per-gene vectors are lastly pooled between genes with the gated attention of (8) to yield the representation $z_A \in \mathbb{R}^{64}$ and per-antibiotic scores $\ell_A \in \mathbb{R}^{7}$.

This design is based on a direct experiment as opposed to reasoning solely. When a convolution is done, only near positions in the input actually are related which is called locality. We checked that this assumption is true throughout the entire pan-genome by training a plain one-dimensional convolutional network on the full sequence of variants,

$$h^{(l)} = \mathrm{MaxPool}\Big(\mathrm{ReLU}\Big(\mathrm{BN}\Big(\mathrm{Conv1d}_l\Big(h^{(l-1)}\Big)\Big)\Big)\Big)\,, \tag{10}$$

with $h^{(0)}$ the embedded variant sequence and four convolutional layers of widths 9, 7, 5 and 3 and channel sizes 64, 96, 128 and 128 in turn, the last of which skips the pooling step and is instead reduced by global average pooling and a single linear readout, and then randomly shuffling the order of the variants. If position were actually meaningful, then shuffling would have hurt performance but it did not: the model performed equally well with the shuffled input, providing further evidence that the order of variants in the pan-genome has no signal and that a convolution at this scale is learning nothing about position. Within a single gene this is not the case, since there the order of sequence is biologically real. The Motif Pattern Encoder thus only uses convolution within blocks of genes, which is precisely where the locality assumption passes the test.

2) Gene Feature Attention Module: The Gene Feature Attention Module reads the presence and absence profile of the genes, a binary vector of 32,441 genes indicating which genes each strain has. It has been employed to detect acquired resistance, but in a totally different manner than is required for setting up a system to detect mutational resistance. The mechanism by which a strain can become resistant is by acquiring a completely new gene, the original strain did not have, usually through a plasmid. This gene is either missing or there it is; so there is no amount of mutation analysis that would find it. The reason why this view is present is to make such a case.

It assigns each gene a learnable embedding, which is just a short vector that the model learns that represents what that gene actually means. This embedding is only permitted to have an effect if the gene is contained in that strain,

$$\tilde{e}_k = x_k\, E_k, \qquad k = 1, \ldots, G_{\mathrm{pa}}, \tag{11}$$

where $E_k \in \mathbb{R}^{d_{\mathrm{token}}}$ is the learned embedding of gene k, and $x_k \in \{0, 1\}$ indicates whether the strain possesses gene k. When $x_k = 0$, $\tilde{e}_k$ is the zero vector, and therefore a missing gene cannot affect the prediction at all regardless of what its embedding has learned. Genes that are absent in a strain are also masked out by setting their attention scores to $-\infty$ just before the softmax of (8), which pushes their weight to exactly zero and redundantly provides the same conclusion.

We cannot apply (8) to $\tilde{e}_k$ directly and this is the only point in the design we alter the mechanism, but not the data it accesses. Gated attention scores every item based on the item representation. But for any gene the strain carries, this representation is $\tilde{e}_k = E_k$, which is the same vector for each strain. The score would be the same for each strain, too, the

softmax would just renormalise based on which genes that strain just happens to have, and the lens would collapse to a global gene ranking based on presence. It was unable to convey the message that a certain gene is important to a certain isolate, which is the whole point of laying focus here. This is not the case with the other two lenses, as the inputs of these lenses already vary depending on the strain.

The score is therefore dependent on the strain. For the remainder of the research, we will refer to this as *strain conditioning*. We ablate it against the standard method in Section V-J.

$$c = \frac{1}{\max(n_b, 1)} \sum_{k=1}^{G_{\mathrm{pa}}} \tilde{e}_k, \qquad n_b = \sum_{k} x_k, \tag{12}$$

Here c is the average embedding of the genes this strain really carries, and each gene is scored in the company it maintains,

$$s_k = w^{\top} \Big(\tanh\big(V \tilde{e}_k + V_c\, c\big) \odot \sigma\big(U \tilde{e}_k + U_c\, c\big) \Big), \tag{13}$$

with V_c and U_c being added projections. A sum of two projections is the same as a single projection on the concatenation $[\tilde{e}_k;\, c]$, but is useful because it broadcasts c rather than realising a tensor of size $B \times 32{,}441 \times 2d_{\mathrm{token}}$, a size that is affordable at this number of genes. The additional cost is two $d_{\mathrm{token}} \times d_{\mathrm{attn}}$ matrices. It increases the neural lenses by 0.035 average PR-AUC under the lineage holdout. The attention weights also modify the genes that are being selected. The attended genes, in turn, contain 41% as much resistance information as the genes selected by SHAP when it is not present. They have 89% as much with it.

The attention calculation is performed at single precision as opposed to half precision, as accumulating the sum over 32,441 genes at reduced precision adds sufficient round-off error to disrupt training. The attended embeddings are pooled and projected,

$$z_B^{\mathrm{pre}} = \sum_{k=1}^{G_{\mathrm{pa}}} a_k\, \tilde{e}_k, \qquad z_B = \mathrm{GELU}(W_B\, z_B^{\mathrm{pre}} + b_B) \in \mathbb{R}^{64}, \tag{14}$$

with per-antibiotic scores $\ell_B \in \mathbb{R}^7$ read off z_B by seven independent linear heads.

In a fully connected layer, each unit would require tens of thousands of weights (one per input), and with a total of only 6,962 training samples, a layer that large would overfit almost without fail. This is avoided by attention which does not increase the number of parameters as the number of genes increases, the same small scoring function is used on each gene separately. It also provides us with something that a dense layer does not: the weights of attention are readable, and we can query which genes the model focused on to make a particular prediction. The interpretability analysis of Pillar 3 is based on that readout.

3) Cross-Antibiotic Context Module: The Cross-Antibiotic Context Module reads the per-gene summary statistics, of shape $(G, 4)$, where each gene is characterized (not by its complete mutation pattern) by four numbers. It does not have the same purpose as the other two lenses: rather than deriving a new type of signal out of the genome, it takes advantage of the interactions among the antibiotics themselves. The rationale is biological. Drug resistance to two drugs of the same class is normally caused by the same mechanism. If a strain develops an enzyme that can hydrolyse cefotaxime, it is likely that it will also hydrolyse ceftazidime, as both are cephalosporins and the enzyme will act on both. In the event that we train seven absolutely independent models, each drug must learn its own mechanism using its own labels, and the drugs that have few resistant samples do not get enough chances to learn anything reliable. When two or more drugs within a drug class have a common representation, evidence from a well-represented drug can assist the model with a poorly-represented drug in the same class. Figure 6 also puts that argument to the test of our own data, and the answer to it is not wholly the one which the biology predicts. That is the shared-mechanism account exactly: the two cephalosporins are by far the most significantly associated pair, at $\phi = 0.84$, and nothing else even remotely approaches that. But below it, membership in the classes ceases to be a useful predictor. Ampicillin complements trimethoprim-sulfamethoxazole at 0.55, in two unrelated classes, as well as it complements amoxicillin-clavulanate at 0.54 within its own; and ciprofloxacin, which has no second member in the present case, is at 0.51 and 0.52 with the two cephalosporins. Resistance genes on the same plasmid or integron travel together, regardless of whether they target the same drug, and this group shows both patterns at once. The figure also shows why small overlap values here cannot be used as a ranking: some antibiotic pairs are measured against only 235 strains (amoxicillin-clavulanate and trimethoprim-sulfamethoxazole), while others are measured against 6,030 (ciprofloxacin and gentamicin), so the values are not directly comparable.

This influences the design that ensues. The cephalosporin connection would be detected by a module that exchanged information solely inside classes (and not between them), and it would fail to learn any cross-class link, which explains why the class-specific heads below are initially succeeded by a refinement step that is free to learn connections not present in the hierarchy.

This is implemented directly in the Cross-Antibiotic Context Module. Each gene's four statistics are first transformed by a small multilayer perceptron (one per gene); then, for global pooling across genes, gated attention is used to compute $z_C \in \mathbb{R}^{64}$. This common representation is then decoupled into one distinct projection for each drug class; the penicillins (AMP, AMC), the cephalosporins (CTX, CAZ), the fluoroquinolone (CIP), the aminoglycoside (GEN), and the folate-pathway inhibitor (SXT). All of the data prior to this split is learned jointly from all seven drugs; all of the data after this is specific to one class,

$$t = \mathrm{GELU}(W_t z_C + b_t), \qquad u_g = \mathrm{GELU}(W_g t + b_g), \tag{15}$$

where $t \in \mathbb{R}^{32}$ is the shared trunk output and $u_g \in \mathbb{R}^{16}$ is the projection for drug class g. Now each antibiotic reads its initial score, according to the projection of its own class only,

$$\ell_{C,a}^{\mathrm{init}} = w_a^{\top} u_{g(a)} + b_a, \tag{16}$$

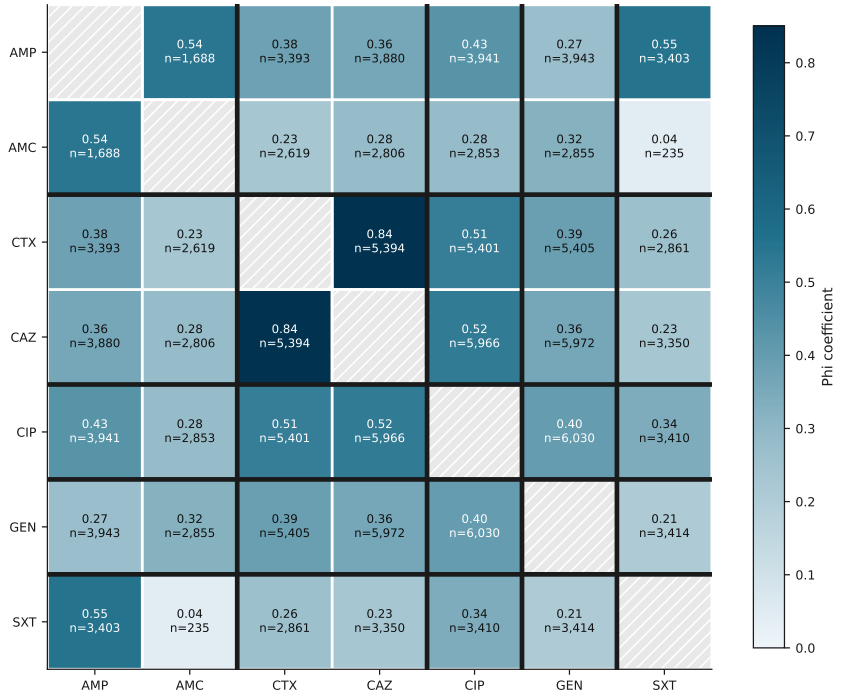


Fig. 6. Pairwise co-resistance (ϕ) between antibiotics, each cell computed on the strains labelled for both drugs (n shown). Lines separate drug classes.

where $g(a)$ is the drug class antibiotic a belongs to. Lastly, a co-resistance refinement step allows the prediction for one antibiotic to modulate the prediction for a related one,

$$r = \mathrm{MLP}\big([z_C\,;\,\ell_C^{\mathrm{init}}]\big)\,, \qquad \ell_C = \ell_C^{\mathrm{init}} + \alpha\, r, \tag{17}$$

with α a learned scalar starting with a value of 0.1. This step is initialised to close to zero, such that the model initially ignores it, and only considers it when the data supports the step which ensures that a spurious relationship between two drugs is not baked in at the beginning.

It is easier to train seven separate models and this is how we started. The issue is the lack of balance in drugs. The number of resistant strains of gentamicin is low compared to ampicillin and therefore there is nothing much to learn by a gentamicin-only model. There is a common base shared across all antibiotics. This base learns from the labels of every drug, not just one. This is exactly what multi-task learning is designed to do. Nonetheless, if the common base acquires a misleading pattern, it influences all drugs. To limit this problem, there is a branch for each drug. Some of those mistakes can still be corrected by these branches.

D. Mixture-of-Experts Gating and Representation Fusion

When each of three lenses has made an opinion about each of the seven antibiotics, their opinions will not necessarily be consistent, and here the question is to what to do with them. The easiest would be to average the three. We discarded that, since averaging supposes that all the three lenses will be equally useful, and they are not. When the resistance of a drug is due to an acquired plasmid gene, the Gene Feature Attention Module should prevail and the Motif Pattern Encoder can do practically nothing with it. In the case of a drug whose resistance is due to a point mutation in a target enzyme, the opposite holds true. A decision based on the average would drown out the voice of a stronger lens with two weaker ones.

The mixture-of-experts gate [11], [12] addresses this by learning weights, rather than fixing them, and by learning them individually, one strain at a time, and each antibiotic at a time. The three representations are combined into a single vector $z = [z_A; z_B; z_C] \in \mathbb{R}^{192}$ and fed through a small network and softmax to obtain

$$g = \mathrm{softmax}\big(\mathrm{MLP}([z_A; z_B; z_C])\big), \quad g \in \mathbb{R}^{B\times 7\times 3}, \tag{18}$$

so that for each antibiotic, the three weights are non-negative and sum to one. The fused score of antibiotic a is then the gate-weighted sum of the lens scores,

$$\ell_a^{\mathrm{fin}} = \sum_{e=1}^{3} g_{a,e}\, \ell_{e,a}, \qquad p_a^{\mathrm{neu}} = \sigma\big(\ell_a^{\mathrm{fin}}\big)\,. \tag{19}$$

One common failure mode of a gate of this type is often referred to in the literature on mixture-of-experts as expert collapse, and is known here as gate collapse. It is often the case that in the early stages of training, one of the lenses is a bit better than the rest, by chance. This slight edge is noticed by the gate and makes it transfer more weight onto that lens. The larger the gradient, the more weight. This means that the lens will learn quicker and even more from the gate. After a few epochs the other two lenses have ceased to be trained at all and the model silently transforms into a single-lens network.

To avoid this we count the spread out of the gate with each antibiotic a by its Shannon entropy,

$$H(g_a) = -\sum_{e=1}^{3} g_{a,e} \log g_{a,e}, \tag{20}$$

which is largest when the three weights $g_{a,e}$ are equal and smallest when the gate has collapsed onto a single lens. Then we introduce an entropy term into the loss, $-\lambda_{\text{ent}}$ times the mean of $H(g_a)$ over the seven antibiotics,

$$\mathcal{L}_{\text{ent}} = -\frac{\lambda_{\text{ent}}}{7} \sum_{a=1}^{7} H(g_a) = \frac{\lambda_{\text{ent}}}{7} \sum_{a=1}^{7} \sum_{e=1}^{3} g_{a,e} \log g_{a,e}, \tag{21}$$

that is minimized with a gate that spreads its weight and maximized with a gate that concentrates its weight on one lens. With a small weight λ_{ent} this pushes the gate to keep all three lenses active whilst leaving it free to favour one where the evidence actually supports it. The symbol is written λ_{ent}, and not β, to keep it distinct from the β of the F_β score in (26). We track the average weight assigned to each lens during training, and automatically increase the entropy weight when any lens's weight goes below a threshold.

E. Stacking Meta-Learner

The decision is eventually made by a logistic-regression meta-learner which combines the neural probability with the five classical out of fold probabilities. This is stacked generalisation in the sense of Wolpert [13], where the predictions of multiple base learners are used as features for a second-level model. Unlike simple averaging, the weighting is not fixed in advance; the six probabilities are treated as inputs to a tiny model that learns from the data how much to trust each of them. That is the difference between consulting six specialists and taking the mean of their replies, and consulting six specialists whose respective track records have been studied with a view to determining which of them is apt to be right in which class of case.

The meta-learner constructs the feature vector ϕ_a of each antibiotic a,

$$\phi_a = \left[p_a^{\text{neu}},\ p_a^{(1)},\ p_a^{(2)},\ p_a^{(3)},\ p_a^{(4)},\ p_a^{(5)} \right], \tag{22}$$

where $p_a^{(m)}$ are the probabilities of the five base learners, and calculates

$$\eta_a = w_a^\top \phi_a + b_a, \qquad \hat{p}_a = \sigma(\eta_a), \tag{23}$$

for each antibiotic, using an L2-penalised logistic regression with $C = 1.0$. It is a conscious decision to train one meta-learner per antibiotic: the drugs are supposed to rely on different models, and having a common meta-learner would push all seven to make one trade-off. The final call is obtained by thresholding the probability,

$$\hat{y}_a = \mathbf{1}[\hat{p}_a > 0.5], \tag{24}$$

with a fixed decision threshold of 0.5. We predict all seven antibiotics jointly, so one forward pass through the model gives a full multi-label resistance profile.

We also tested a neural meta-learner. It is a more complex combination of the six base predictions. The first step uses a small feed-forward layer to generate one token per antibiotic. Subsequently, a four-head self-attention layer allows the seven tokens to exchange information. Then, the final score of each drug is provided by a separate decoding head. This was aimed at capturing co-resistance profiles, including the cefotaxime-ceftazidime relationship. This drug-to-drug relationship cannot be learnt by a separate linear model. The neural model, however, did not fare well. It scored 0.825 against the simple meta-learner's 0.849. It was not the number of input features that was the problem. The neural embedding is folded into the meta-feature vector, bringing it to 170 dimensions. Nevertheless, there are only about 5,000 strains behind those numbers. The model is further complicated by the attention layer and seven decoding heads. Consequently, it can easily fit noise in such a small dataset. We can directly read the weights of the linear meta-learner, letting us see more explicitly than a deeper meta-learner might allow how the final decision broke down in terms of how much to trust the neural arm and how much to trust each classical model for every antibiotic.

F. Implementation Details

All of the settings below are global across the model. Most follow traditional practice and are therefore included for completeness; a few are less obvious and are explained, where necessary.

1) Binary Classification Setup: In the case of each antibiotic, the model aims to predict either resistance (labelled 1) or susceptibility (labelled 0). The inputs are the three genomic views discussed in the top-level overview and each raw score is transformed into a probability using the logistic sigmoid from (1). All seven antibiotics are predicted in the same forward pass, making it a multi-label problem instead of a multi-class problem: a strain may be resistant to multiple drugs simultaneously and the predictions do not have to compete with each other.

2) Handling Class Imbalance: Susceptible strains by far outnumber resistant ones for various drugs, which poses a very real trap: a model predicting susceptible in all cases scores highly on accuracy and is useless, but which is approximated well by a model that predicts the majority class as the easiest way to minimize the loss. To avoid this, the resistant and susceptible samples are reweighted in inverse proportion to their frequency of occurrence, such that a mistake in getting a rare resistant strain is much costlier. In the classical tree-based learners it can be accomplished by the `scale_pos_weight` parameter,

$$\texttt{scale_pos_weight} = \frac{N_{\text{neg}}}{N_{\text{pos}}}, \tag{25}$$

where N_{neg} and N_{pos} are the numbers of susceptible and resistant samples. In Logistic Regression and Random Forest, this is accomplished by using balanced class weights, and in the neural lenses, by using positive class weighting in the loss. For this reason, we use precision-recall AUC and not accuracy during the evaluation as accuracy would conceal precisely the failure we want to avoid.

3) Threshold Selection: To arrive at the final decision, we use a constant probability threshold of 0.5 for each of the antibiotics, as in (24). This is the indifferent option. It places the same emphasis on the two kinds of errors. But in clinical practice this is not the case. The issue of predicting a resistant strain as being susceptible can result in administering a drug that is not effective. The reverse error is only going to result in picking another drug. A recall-oriented adjustment was made according to the F_β score,

$$F_\beta = \frac{(1+\beta^2)\cdot \text{Precision} \times \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}, \tag{26}$$

with $\beta = 2$ to give four times as much weight to recall as to precision, would lower the threshold and capture more resistant strains while increasing the number of false alarms. To avoid tuning the numbers in our favour, all headline results in this research have been reported at the neutral threshold of 0.5 and then per-antibiotic threshold tuning has been directly tested in Section V, fitting the threshold on validation and applying it to test.

4) Optimization and Training Schedule: The training of the neural lenses, the gate and the fusion step is done in a simultaneous way, so that each lens learns to be useful in conjunction with the others and not optimised alone. The objective consists of three terms: a binary cross-entropy loss over the fused scores, the entropy regulariser in (21), and an auxiliary loss on each lens. This extra loss term is important. In its absence, a lens which is momentarily disregarded by the gate achieves extremely limited gradient. This leads to its cessation of improvement. This problem is avoided by the auxiliary loss. It aids all lenses in continuing to learn, even when they are not being selected by the gate. Optimization employs the AdamW optimiser that keeps the weight decay and gradient update decoupled, and the learning rate of 10^{-3}, annealed on a cosine schedule over the epoch budget, weight decay of 10^{-4}, batch size of 16, gradient clipping of norm 1.0, and mixed-precision training with the single exception described in Section IV-C2, where the attention sum over 32,441 genes is kept at single precision, with up to 50 epochs and early stopping after 8 consecutive no-improvement epochs of validation mean precision-recall AUC. The inclusion of gradient clipping is because, when attention is placed over 32,441 genes, the gradients can sometimes be quite large, causing the training to become unstable. The five classical base learners are trained individually via 5-fold cross validation, with out-of-fold predictions, which are then used to fit the logistic-regression meta-learner, along with the neural probabilities. The most important hyperparameters are summarised in Table VIII.

TABLE VIII
HYPERPARAMETER CONFIGURATION FOR THE CLASSICAL BASE LEARNERS AND THE NEURAL MODEL.

Component	Key hyperparameters
XGBoost (gene)	n_estimators=1500; max_depth=5; min_child_weight=5; learning_rate=0.05; subsample=0.8; colsample_bytree=0.7; reg_alpha=0.1; reg_lambda=1.0; gamma=0.1; tree_method='hist'; early_stopping_rounds=150
XGBoost (raw)	χ^2-prefilter (top 50,000 SNPs); n_estimators=1500; max_depth=5; learning_rate=0.05; subsample=0.8; colsample_bynode=0.5; reg_alpha=0.1; reg_lambda=1.0; early_stopping_rounds=150
LightGBM	n_estimators=1500; num_leaves=63; max_depth=6; min_child_samples=20; learning_rate=0.05
Random forest	n_estimators=200; max_depth=12; min_samples_leaf=10
Neural lenses and gate	AdamW; learning_rate=10^{-3} with cosine annealing; weight_decay=10^{-4}; batch_size=16; up to 50 epochs; early stopping patience 8; gradient clipping at norm 1.0; entropy weight $\lambda_{\text{ent}} = 0.01$
Meta-learner	Logistic regression (L_2, $C = 1.0$), one per antibiotic, fitted on 5-fold out-of-fold inputs

V. RESULTS

This section outlines what our models produced and what those numbers mean. First we describe the measures that we used in evaluating performance and the results of the proposed model, and then decompose the model, layer upon layer, to see where its performance actually comes from. The remainder turns to the evidence that matters most for this research's central claim: how the model behaves under a genuine lineage holdout, on a separate collection, and when the genes it claims to rely on are put to the test.

A. Performance Evaluation

The smaller group are the resistant strains, and even for a single antibiotic only about one in five strains is resistant, so plain accuracy would not reveal the problem: a model that declared every strain susceptible would still seem accurate, but would be useless in a clinic. The primary metric that we are interested in is the area under the precision–recall curve (PR-AUC), which captures the performance of a model in terms of precision-recall at all threshold levels and does not encourage a model to make majority class predictions. It is defined as

$$\text{PR-AUC} = \sum_n (R_n - R_{n-1})\, P_n \tag{27}$$

where P_n and R_n are the precision and recall at the n-th threshold.

We also report a few other measures, since there is no single number which tells the entire story. Matthews Correlation Coefficient (MCC) employs all four cells of the confusion matrix simultaneously, and remains truthful in case of imbalanced classes:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{28}$$

MCC can be between -1 and $+1$ (where 0 indicates model guessing), and all values are calculated per antibiotic using a one-versus-rest approach. We also report the two members of the F_β family of (26) that this work uses, obtained by setting $\beta = 1$ and $\beta = 2$ in that expression: F_1 is the harmonic

TABLE IX
TRILENS-MOE-STACK ON THE RANDOM SPLIT, PER ANTIBIOTIC.

Drug	PR-AUC	MCC	F1	F2	Acc.
CIP	**0.9773**	0.9088	0.9282	0.9197	0.9692
AMP	0.9440	0.7866	0.8932	0.8917	0.8933
CTX	0.9019	0.7904	0.8173	0.8126	0.9530
SXT	0.8352	0.6954	0.7875	0.7730	0.8707
CAZ	0.8299	0.6928	0.7290	0.7169	0.9356
AMC	0.7482	0.5237	0.6563	0.6250	0.8035
GEN	**0.7037**	0.6095	0.6386	0.5915	0.9341
Average	**0.8486**	**0.7153**	**0.7786**	**0.7615**	**0.9085**

Note: Ciprofloxacin has the highest PR-AUC (0.9773) and gentamicin the lowest (0.7037), the spread the text below discusses; the last row averages across all seven antibiotics. Bold marks the extreme values in a column, best and worst alike.

TABLE X
CLINICAL ERROR RATES AT A THRESHOLD OF 0.5, ORDERED BY VERY MAJOR ERROR RATE.

Drug	VME	ME	CA	Meets CLSI
CIP	**0.086**	0.015	0.969	No (ME only)
AMP	0.109	0.104	0.893	No
CTX	0.190	0.026	0.953	No (ME only)
SXT	0.236	0.080	0.871	No
CAZ	0.291	0.033	0.936	No
AMC	0.394	0.108	0.804	No
GEN	**0.436**	0.023	0.934	No (ME only)

Note: Ciprofloxacin has the lowest very major error (VME) rate at 0.086 and gentamicin the highest at 0.436. No antibiotic meets both CLSI limits at once; "No (ME only)" means only the major-error limit is met.

mean of precision and recall, and F_2 weights recall four times higher, since failing to identify a drug-resistant strain is worse than falsely labelling a susceptible strain resistant.

We show accuracy for thoroughness, but we do not use it. Measures of machine learning are inadequate by themselves to tell a clinician whether a tool is safe, and so we also report the microbiology vocabulary. A *very major error* (VME) is a resistant strain called susceptible, which is the dangerous case; a *major error* (ME) is the reverse, which costs a drug option but does not put the patient directly at risk; and *categorical agreement* (CA) is the proportion of strains on which our call matched the laboratory result. All these evaluations determine three things: the degree to which the model ranks strains well (PR-AUC), the degree to which it makes balanced decisions at some threshold (MCC, F1, F2), and whether its errors would be tolerated in a hospital (VME, ME, CA).

The average PR-AUC is 0.849, yet the average conceals a broad distribution, and the interesting part is the spread. The clearest case is ciprofloxacin, at 0.9773 with MCC 0.9088, which is close to a practical ceiling for this kind of data, followed by ampicillin at 0.9440. On the other hand, gentamicin comes in at 0.7037 and amoxicillin-clavulanate at 0.7482, meaning the spread of best to worst is nearly 0.27 for models trained of exactly the same genomes with the exact same architecture.

The accuracy column is different from the others. Gentamicin has the lowest PR-AUC value, but high accuracy value of 0.9341 as the proportion of strains resistant to gentamicin is small, so a model can achieve a high accuracy, but fail to detect many resistant strains. This is the same trap as above, and the reason accuracy is not our primary measure. As far as a hospital is concerned, it is the type of error that the model produces, of which the results are documented in Table X.

Ciprofloxacin reappears as the strongest, having a VME of 0.086, and a categorical agreement of 0.969. Gentamicin is weakest with 0.436 (almost half of all resistant strains would be reported sensitive) and amoxicillin-clavulanate is almost as poor (0.394). The ordering is important, as an almost identical ordering is rediscovered under the lineage holdout in Section V-E in an entirely different way.

The categorical agreement column is the same as the accuracy column of Table IX, and this is not by chance. During data preparation the intermediate results were grouped with the resistant results, so the intermediate category no longer exists in this dataset. Categorical agreement therefore reduces to binary accuracy here. Gentamicin illustrates this point clearly, with a categorical agreement of 0.934 versus a VME of 0.436. On the same strains, the same model is seen to be strong by one measure and unusable by the other.

The CLSI limits require both the VME and ME rates to stay under three percent [9]. Three antibiotics satisfy the major error limit, which are ciprofloxacin at 0.015, gentamicin at 0.023 and cefotaxime at 0.026 but none satisfy the very major error limit, therefore, the model is not yet prepared to be used as a diagnostic test.

Two qualifications apply to that comparison, and the second is the more serious of them. The first is that the limits are written for the verification of a commercial susceptibility testing system against a challenge panel that is deliberately enriched for resistant and near-breakpoint isolates, whereas we apply them to a retrospective genomic model on a convenience collection; a passing score here would not constitute regulatory evidence, and we use the limits only because they are the closest published statement of what error rate a laboratory will tolerate. The second is that we collapsed the intermediate category into resistant, so our three rates are computed on a two-category comparison, while the standard defines them on the three-category susceptible, intermediate and resistant comparison and excludes certain intermediate-involving discrepancies from the counts altogether. Collapsing in this direction moves genuinely intermediate strains into the resistant class, which can only raise our apparent very major error rate relative to the standard definition and leaves the major error rate essentially unaffected. The direction of the bias is therefore against us and the conclusion that no antibiotic clears the very major error limit is safe, but the individual rates are not the same quantity the standard tabulates and should not be quoted as though they were.

The obvious response is that the fixed 0.5 threshold was never tuned to clinical cost, and that the numbers above could be an artefact of positioning the cut. To do this, we trained a threshold for each antibiotic on the validation set, constraining the F_2 objective of (26) and capping the major error rate, and assessed each on the test set. While the tuning for F_2 moves the mean very major error rate only from 0.249 to 0.210, still

TABLE XI
BOOTSTRAP CONFIDENCE INTERVALS FOR THE PROPOSED MODEL ON THE RANDOM-SPLIT TEST SET, FROM 1,000 RESAMPLES.

Drug	PR-AUC	95% confidence interval
AMP	0.9440	[0.919, 0.965]
CIP	0.9773	**[0.961, 0.990]**
AMC	0.7482	[0.680, 0.809]
SXT	0.8352	[0.773, 0.891]
GEN	0.7037	**[0.612, 0.786]**
CTX	0.9019	[0.851, 0.943]
CAZ	0.8299	[0.762, 0.882]
Average	**0.8486**	*not defined*

Note: Ciprofloxacin's interval, [0.961, 0.990], is the tightest in the table, backed by the most labelled data; gentamicin's, [0.612, 0.786], is the widest, backed by the least. No interval is given for the average, because the seven per-drug resamples are drawn from overlapping strain sets and their intervals are therefore not independent; the average is reported as a point summary only. Bold marks the extreme values in a column, best and worst alike.

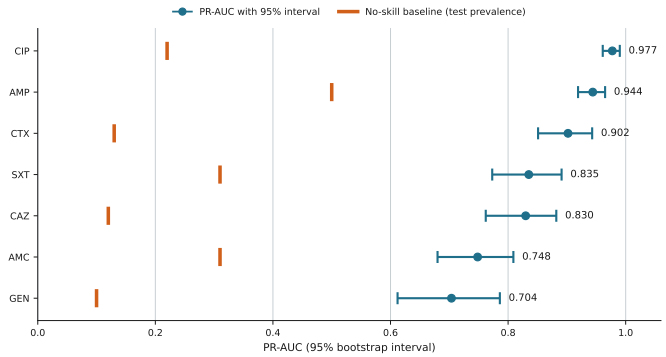


Fig. 7. PR-AUC with 95% bootstrap intervals, against each antibiotic's no-skill baseline (test-set prevalence).

far above the limit of 0.03, and forcing the major error rate to remain inside the limit pushes the mean very major error rate to 0.437, neither rescues the model. In fact, none of the antibiotics was able to achieve both thresholds at any threshold value that we tested, because shifting the cut trades one kind of error against the other instead of cutting down on both. It is not the position of the cut that drives the clinical shortfall, however, but the model's ability to separate the two classes. As such, future work should not focus on calibration, but on the feature representation.

B. Statistical Analysis

In our test set there are 1,045 strains and in some antibiotics there are only a few hundred of known label, so one or two points difference can be simply accidental. We resampled the test set (1,000 bootstrap resamples with replacement) to recalculate PR-AUC, and used the 2.5th and 97.5th percentiles as a 95% confidence interval. The result is shown in Table XI and plotted in Fig. 7 as the intervals against the no-skill baseline of the individual drugs, i.e., the range within which the raw scores are most misleading: a PR-AUC has to be read against the prevalence of the drug it represents, not against the other drugs in the column. The PR-AUC of ampicillin, 0.944, sits above a no-skill baseline of roughly one resistant strain in two. The score of gentamicin, 0.7037, appears smaller in absolute terms, but its no-skill baseline is only about one resistant strain in ten. Gentamicin's raw score is lower, yet it is the greater gain over chance.

Gentamicin ranges 0.612 to 0.786, a range of nearly 0.18, and this is simply the result of few resistant strains, but ciprofloxacin, which has more labelled strains and a stronger signal, has a much smaller range. Any difference between two models on this test set must therefore exceed this interval width before it can be termed significant. None of the differences reported in this section between closely related methods clears that bar.

One limit of these intervals should be stated plainly, because it governs how every architectural comparison in this section may be read. A bootstrap over the test set quantifies the variability that comes from which strains happen to fall in the test partition. It does not quantify the variability that comes from training itself. Every model reported here was trained once, under the single fixed seed of Section III-G, so the effect of initialisation, of batch ordering and of the stochastic elements of the classical learners is not represented in any interval we report. Differences of the order of a few thousandths of PR-AUC, such as the $+0.009$ that separates the full model from the strongest single classical learner in Section V-C or the $+0.0024$ that strain conditioning buys on the random split in Section V-J, are therefore of a size that seed variation alone could plausibly produce, and we do not claim them as real effects. We treat as established only those differences that are large relative to the interval widths above, namely the gap between the random split and the lineage holdout and the gap between the two groups of antibiotics. The $+0.035$ that strain conditioning delivers to the neural lenses alone under the holdout sits between the two cases: it is an order of magnitude larger than the same term's effect on the random split, and it moves three interpretability measures with it, which is more than a seed artefact would be expected to do, but on its own it does not clear the interval widths either. Repeating the full protocol over several seeds, and reporting a mean and standard deviation for each cell, is the natural next step and is the first item of future work in Section VII.

A second robustness check belongs with these intervals. Some bacterial collections have nearly identical strains. If one of the copies ends up in training and the other ends up in the test set, the model may score well on this task simply by recognizing something it has already seen, so we checked for this reason. To assess this, we calculated the Hamming distance from each test strain to its nearest training strain. The closest pair differed by only 0.26% of their SNP positions, and, out of 1,045 test strains, 47 were within 0.5% of a training strain, which are considered "effectively duplicates." The 0.5% cut is a working threshold and not a standard one: on the 535,439-position matrix it corresponds to roughly 2,700 differing positions, which is an order of magnitude below the typical separation of two strains of the same sequence type in this collection, so pairs inside it are far closer than clonality alone would explain. We report the sensitivity of the result to that choice below rather than resting on the cut itself.

Then, we removed these 47 strains, and scored the model

TABLE XII
COMPONENT ABLATION ON THE RANDOM SPLIT, WITH THE PR-AUC GAIN EACH ADDED PART CONTRIBUTES OVER THE STAGE BEFORE IT.

Stage	AMP	CIP	AMC	SXT	GEN	CTX	CAZ	Avg	Gain
Stage 1: lenses only	0.8920	0.9248	0.6249	0.7217	0.5168	0.8103	0.7267	0.7453	N/A
Stage 2: + MoE gate	0.8970	0.9489	0.6347	0.7453	0.5835	0.8659	0.7893	0.7807	+0.0353
Stage 3: + stacking	0.9440	0.9773	0.7482	0.8352	0.7037	0.9019	0.8299	**0.8486**	+0.0679

Note: The full model reaches an average PR-AUC of 0.8486; classical stacking adds +0.0679 over stage 2, about twice the gate's own +0.0353 contribution.

anew on the remainder. PR-AUC decreased on average by 0.0041, with ceftazidime falling by the largest, of 0.0209, and amoxicillin-clavulanate increasing by 0.0075. It is very slight and not directional, which informs us that near-duplicates will not be the primary factor that supports our findings. Nonetheless, it remains a real effect and reporting the deduplicated number provides a truthful lower bound.

C. Component Ablation

Each neural lens reads a different part of the genome and a gate determines the trustworthiness of each lens, with five classical learners combined by a stacking layer. To determine whether each part is warranted, we removed each component in turn and measured the impact on performance. We trained three variants using the same data, seed and test set. **Stage 1** is the three lenses only; they average their outputs with equal weighting and no gate is used. **Stage 2** is the three lenses and the mixture-of-experts gate, such that the model learns the extent to which to rely on individual lenses on individual samples and antibiotics; this is the neural part of the model on its own. **Stage 3** is the complete model in which the neural output is appended with the five classical base learners via the meta-learner. Table XII shows the result.

There are two points that are established. First, there is no dead weight: for each and every antibiotic, the performance improves when it goes from Stage 1 to Stage 2 and then to Stage 3. Second, the parts do not aid equally. Combining neural predictions with classical learners achieves on average about 0.0679 more than the neural design alone, while the gate achieves only 0.0353. The neural lenses are useful, but not sufficient by themselves, and thus the final model is best referred to as a hybrid rather than as a neural model.

The average weight assigned to each lens by the gate is shown in Table XIII, and the weights by antibiotic in Fig. 8.

The trend is the same. Most of the weight of each antibiotic, 0.590 to 0.900, is in Lens B, which reads which genes are present or absent, whereas Lens A, the convolutional lens reading mutation patterns within genes, never goes above 0.098. At least this is consistent with the biology as a large part of the resistance in *E. coli* is conferred by acquired genes.

Ciprofloxacin is where the table stops agreeing with the design. The rationale for the gate in Section IV-D is a prediction: a drug whose resistance is based on an acquired gene should route to Lens B, whereas a drug whose resistance is based on a point mutation in a target enzyme should route to Lens A. Ciprofloxacin is the most obvious point-mutation drug that we have in our panel, and the one that our SHAP analysis suggests is related to *gyrA* and *parC*; yet Lens A receives 0.035 for this drug, the lowest value in the table. The prediction fails on the one antibiotic where it was most testable. This is the highest value for Lens C, but it is difficult to read from either direction. The lens shares drug-class head evidence, but ciprofloxacin is the only fluoroquinolone in the panel and has no same-class partner, and although its co-resistance refinement step may still link ciprofloxacin to ceftazidime, which shares $\phi = 0.52$ in this collection (Fig. 6), that is co-occurrence rather than mechanism. On neither reading can 0.374 be taken as the gate finding the mechanism.

The gate therefore tells us which lens the model relies on, which is worth knowing, but its routing is not in the direction the design assumed. The same conclusion is drawn in the attention audit in Section V-I. A second reading of the table points the same way. In the final weights, trimethoprim-sulfamethoxazole reaches 0.900. Three more drugs sit above 0.81: ampicillin at 0.852, ceftazidime at 0.835, and cefotaxime at 0.815. It seems that the penalty would have slowed, but not stopped, collapse, and therefore, these weights are possibly

TABLE XIII
AVERAGE GATE WEIGHTS ON THE TEST SET. EACH ROW SUMS TO ONE.

Drug	Lens A	Lens B	Lens C
AMP	0.070	0.852	0.078
CIP	**0.035**	0.590	0.374
AMC	0.095	0.742	0.163
SXT	0.052	**0.900**	0.047
GEN	0.098	0.731	0.171
CTX	0.064	0.815	0.121
CAZ	0.044	0.835	0.122

Note: Ciprofloxacin's Lens A weight of 0.035 is the lowest in the table despite being the drug that lens was built for, and trimethoprim-sulfamethoxazole's Lens B weight of 0.900 is the highest, closest to gate collapse; both are discussed below.

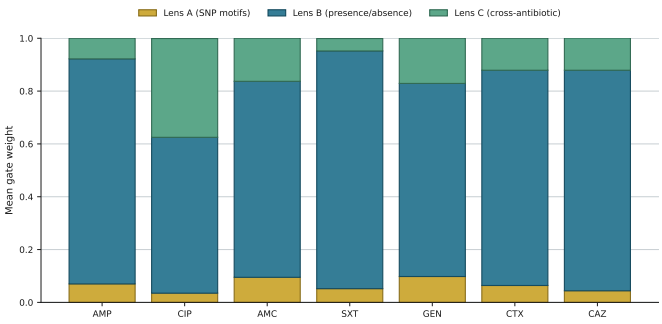


Fig. 8. Mean gate weight assigned to each lens, by antibiotic, on the test set. The three weights sum to one for each antibiotic; these are the values of Table XIII drawn as stacked bars.

TABLE XIV
COMPARISONS OF THE PROPOSED MODEL WITH EXTERNAL BASELINES AND RESULTS OF THE COMPONENT MODELS. THE PHYLOGROUP-ONLY AND PCA BASELINES WERE NOT EVALUATED WITH THE LINEAGE HOLDOUT.

Model	Random split	Lineage holdout
External baselines		
Phylogroup label only	0.2914	not run
TabNet	0.6951	0.3255
XGBoost on 1,024 PCA components	0.8033	not run
Components of the proposed system		
Random Forest, gene features	0.7809	0.4931
Logistic Regression, gene features	0.7923	0.4408
LightGBM, gene features	0.8338	0.5516
XGBoost, gene features	0.8375	0.5507
XGBoost, χ^2-filtered SNPs	0.8400	0.6299
TriLens-MoE-Stack	**0.8486**	**0.6337**

Note: The last row is the proposed model, the one every other row in the table is being measured against.

associated with optimisation-like dynamics as well as with biology.

Table XIV distinguishes between two similar but different questions. The first is the comparison of the proposed model with external models. There are three external baselines: a null model using only the phylogroup label, TabNet and the gradient-boosting model using principal component analysis (PCA) components that this project began with. The second question is what each component of the proposed system contributes to the overall result. With this in mind, the five base learners are tested individually as well. But these are not external baselines. The proposed system consists of multiple learners, each feeding the meta-learner. Thus, comparing these learners with the full system is not an external comparison, but rather an ablation. All models were evaluated with the same data splits and the same five-fold protocol. The values reported are the mean PR-AUC scores of the seven antibiotics.

On the random split the model produces the highest score of all entries in the table. If only the eight-way phylogroup label is used as a baseline, the PR-AUC reaches just 0.2914, against a mean resistance frequency across the seven drugs of 0.250, which is effectively the floor for this task. Coarse lineage membership therefore buys about 0.04 over chance and no more. The specialized tabular deep-learning baseline, TabNet, achieves only 0.6951, behind every single classical learner, and falls to 0.3255 under the lineage holdout, where even the weakest classical learner still reaches 0.4408: added depth does not help on this problem under either test. The five classical learners themselves range from PR-AUC 0.7809 to 0.8400, with the three gradient-boosting models on the higher end of that span. The highest entry in the table, at 0.8486, is TriLens-MoE-Stack, which leads every external baseline on the average and is ahead of its own strongest component on five of the seven antibiotics.

It is not just the direction of the result that matters, but also the size of the improvement. The proposed model outperforms the PCA baseline by 0.0453, TabNet by 0.1535 and the null model by 0.5572. The PCA comparison is the most telling as it demonstrates what this project contributed, since this project began with the PCA model. For individual antibiotics, the proposed model has scores above the upper limit of the baseline interval for ampicillin, ciprofloxacin and trimethoprim-sulfamethoxazole. Its score is within the interval for the other four antibiotics.

D. Comparison with Published E. coli Studies

Table XV places this work beside the published *E. coli* studies of Table I, restricted to those working on a comparable collection and task. The headline figures in the fourth column are not directly comparable with one another, because the collections, the drug panels and the reported measures all differ, and Section V-E shows that a score cannot be read without the prevalence behind it. The column that carries the argument is the fifth: of the five earlier studies, one withholds a lineage and reports the result only as a supplementary analysis, one leaves an MLST-blocked split available as an option it does not use for its headline number, and three do not test on an unseen lineage at all. None corrects for the prevalence shift the holdout produces.

E. The Lineage Holdout

The question posed in this subsection is more difficult: does the model continue to work when it is presented with a strain it has never seen before? The phylogroup-stratified split fails to do it: it rearranges the information within eight large clusters, thus leaving a strain and a close one on different sides, and its performance under it changes little, between 0.849 and 0.853. This is not a test of unfamiliar strains, and therefore could not have indicated a drop in any way.

The holdout which answers it makes use of the type of sequence of each strain, a common designation of bacterial genetics according to which strains are clustered by lineage. We successfully called a type of sequence to 6,487 of the 6,962 strains, or 93.2%, with a total of 458 types, including a couple of giant ones: sequence type 73 alone has 884 strains, or nearly one in every eight of the samples. There were two rules prior to peeking at any outcome. Any strain whose sequence type had no reliable sequence type was put into training and here, any sequence type that comprised over five percent of the data was also put into training, as a test set constructed using mostly a single clone would not be testing generalisation. That condition was fulfilled by four types, namely, ST73, ST131, ST95 and ST69. The other 454 sequences were assigned whole (not split), using a fixed random seed such that no sequence type appears in more than one of the training, validation and test sets.

The outcome was just the opposite. Mean performance plummets to 0.634 when they are split by the lineage holdout, a decrease of over a fifth, and that is what the phylogeny split had been unable to discover since it did not wipe any lineage off training. Table XVI positions the three splits next to each other.

The trend is not anywhere near homogenous and Fig. 9 renders the split apparent: three antibiotics struggle to move, four fall sharply. Ampicillin has a loss of only 0.040 and gentamicin has a loss of 0.458. It is the same ranking as in Sections V-A and V-C, only much more emphatic here, since

TABLE XV
THIS WORK AGAINST PUBLISHED *E. coli* RESISTANCE-PREDICTION STUDIES DRAWN FROM TABLE I. THE REPORTED-RESULT COLUMN IS QUOTED AS EACH STUDY STATES IT AND THE ENTRIES ARE NOT DIRECTLY COMPARABLE, AS THE COLLECTIONS, DRUG PANELS AND MEASURES DIFFER; THE FINAL COLUMN RECORDS WHETHER THE STUDY TESTS ON A LINEAGE IT NEVER TRAINED ON.

Study	Collection and drugs	Evaluation split	Reported result	Tested on an unseen lineage?
Moradigaravand et al. (2018) [24]	1,936 *E. coli*, 11 compounds	Random, with an ST131 lineage holdout as a supplementary analysis	Gradient boosting best of the three learners	Yes, but only as a supplementary analysis; the ST131 holdout scores about 0.28 below a size-matched random control
Her and Wu (2018) [26]	59 *E. coli*, 12 drugs	Leave-one-out cross-validation	Accessory genome carried most signal	No
Ren et al. (2022) [27]	Giessen (987) and public (1,509) *E. coli*	Stratified cross-validation with an independent external set	RF best, AUC around 0.96	No
Orcales et al. (2024) [25]	1,936 *E. coli*	Cross-validation, with an MLST-blocked option	XGBoost most reliable	Available as an option, not used for the headline result
Siddiqui and Tarannum (2026)† [35]	Giessen (809) *E. coli*, 4 antibiotics	Stratified 80/20 holdout	MCC 0.926 for CIP, macro F1 0.691 for GEN	No
This work	**6,962 *E. coli*, 7 antibiotics**	**Random, phylogroup-stratified and sequence-type lineage holdout**	**PR-AUC 0.849 on the random split**	**Yes, as the primary design: 0.634 as measured and 0.723 once the test set is resampled to the training prevalence**

Note: The last row is this work. Bold marks that row and not a best result, because the fourth column is not a like-for-like comparison. †Prior work of one of the present authors.

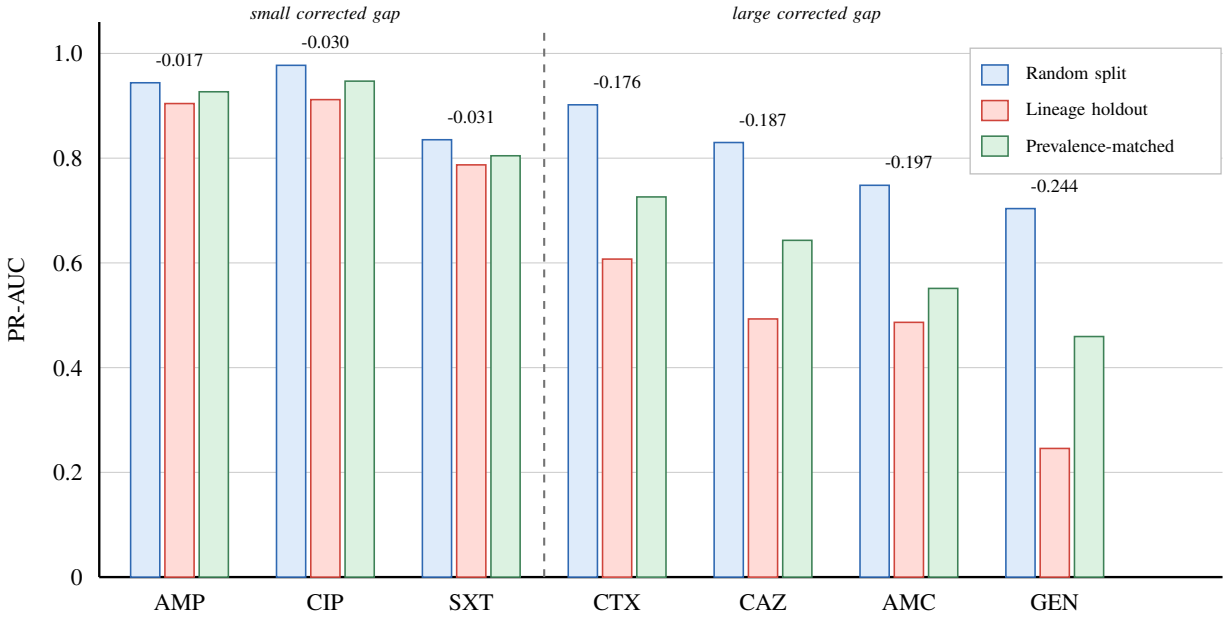


Fig. 9. Random split, the sequence-type lineage holdout as measured, and the same holdout after the test set is resampled to the training prevalence. Bars are ordered by the prevalence-corrected gap, which is printed above each group. Three antibiotics lose 0.017 to 0.031 once prevalence is corrected for and four lose 0.176 to 0.244, with nothing in between. The values behind this figure are in Table XVII.

this is the first time a lineage has been removed as opposed to reshuffled. The third bar in each group takes the prevalence effect out, and the rest of this subsection explains what that means.

There are two objections to be met before all this can be credited. One is that a failure to generalise is not the only way a drop this large can occur, it can also be caused by near-duplicate strains. On this split zero of the 1,045 test strains had a close match in training, using the same nearest-neighbour scan described in Section V-B, which found 47 such strains on the random split. The middle column of the table demonstrates this for the phylogroup-stratified split: four antibiotics outperform the random split and none declines by more than 0.022, which is not the behaviour of a split that truly withholds a lineage. The second objection does need correcting for: the holdout downwardly shifts the test-set resistance prevalence for each antibiotic by between 2.5 and 12.1 percentage points; gentamicin has only 38 resistant strains left. That has to be distinguished from the drop before the drop can be interpreted as a loss of skill.

An unskilled classifier has an expected PR-AUC equal to the prevalence of the positive class. This means that the whole

TABLE XVI
PER-ANTIBIOTIC PR-AUC UNDER ALL THREE SPLITS, WITH THE DROP MEASURED FROM THE RANDOM SPLIT TO THE LINEAGE HOLDOUT.

Drug	Random	Phylogroup	Holdout	Drop
AMP	0.9440	0.9338	0.9043	**−0.040**
SXT	0.8352	0.8373	0.7872	−0.048
CIP	0.9773	0.9617	0.9118	−0.066
AMC	0.7482	0.7963	0.4865	−0.262
CTX	0.9019	0.8807	0.6073	−0.295
CAZ	0.8299	0.8481	0.4930	−0.337
GEN	0.7037	0.7136	0.2457	**−0.458**
Average	**0.8486**	**0.8531**	**0.6337**	

Note: Ampicillin's drop of −0.040 is the smallest under the lineage holdout and gentamicin's −0.458 the largest, the two-group pattern this section keeps coming back to. Bold marks the extreme values in a column, best and worst alike.

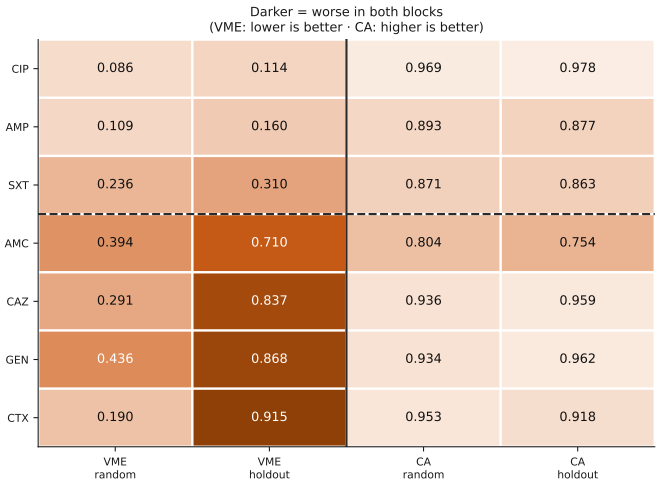


Fig. 10. Very major error rate (left pair, lower is better) and categorical agreement (right pair, higher is better), random split against lineage holdout. Drugs are ordered by holdout VME; the dashed line marks the split between the two groups. Each cell is annotated with its value, so the figure can be read without reference to the colour scale.

scale on which PR-AUC is judged moves with prevalence. A test partition containing fewer resistant strains will therefore score lower even if the model ranks the strains in the same order as before. We refer to this as the no-skill baseline rather than a floor, because a poor enough classifier can still score below it. The opposite extreme is gentamicin, with only 38 resistant in its holdout test set, a prevalence of 4.2% as compared to 14.6% in training.

We directly measured the size of this effect. We repeatedly subsampled the susceptible strains in the holdout test set to obtain the same prevalence as the training set and recomputed PR-AUC on the same saved predictions for each antibiotic. No model was re-trained and no resistant strain was abandoned. Table XVII presents mean over 2,000 resampling rounds with 95% interval and area under the receiver operating characteristic curve (ROC-AUC) with no-skill value of 0.5 irrespective of prevalence.

Only about two-fifths of the headline drop is a prevalence effect. Average PR-AUC falls from 0.8486 (random split) to 0.7226 when prevalence is fixed, a difference of 0.126, instead of 0.6337, a difference of 0.215, as the raw holdout values would have indicated. As its own ROC-AUC of 0.7426 already indicated, gentamicin is the one that most recovers, and its PR-AUC increases from 0.2457 to 0.4594: the model had not ceased to rank resistant strains above susceptible strains, it has merely been forced to do so on a test set where only one in twenty-four strains is resistant.

Adjusting the prevalence does not remove the discovery, instead it clarifies it. The seven antibiotics remain divided into the same two categories, but even more distant relative to each other. Ampicillin, ciprofloxacin and trimethoprim-sulfamethoxazole have a loss of 0.017 to 0.031 in PR-AUC, which is near nothing, and cefotaxime, ceftazidime, amoxicillin-clavulanate and gentamicin have a loss of 0.176 to 0.244, and no antibiotic is in between. A model that had merely deteriorated on a more challenging set of tests would exhibit an even distribution of losses among all seven drugs rather than this two group divide. Amoxicillin-clavulanate is the cleanest: its prevalence changes only by 5.9 points, thus it recovers the least of the four hard drugs (a PR-AUC increase of 0.065, compared to gentamicin's 0.214) and also has the lowest ROC-AUC of all seven, at 0.6894, so the least of its loss can be attributed to prevalence at all.

Two caveats apply to this table. First, gentamicin's prevalence-matched interval is wide, [0.398, 0.511], because it is anchored by only 38 resistant strains, so its exact value should be read with caution even though its direction is clear. Second, the holdout prevalence used for matching is the *training* prevalence, which is slightly higher than the random-split test set's prevalence; this makes the comparison somewhat generous to the holdout, so the true gap may be a little larger than the 0.126 reported above.

F. Clinical Cost of the Lineage Holdout

What the holdout costs in clinical terms is the next question. Section V-A reported clinical error rates on the random split. Of greater interest is what happens to the errors that get to a patient if the strain is from an unfamiliar lineage. Table XVIII does this, using the same saved predictions and the same fixed threshold of 0.5, and Fig. 10 does the same with the numbers arranged by severity.

The average very major error rate increases from 0.249 to 0.559 and for three antibiotics, the model has completely ceased to detect resistance. It detects 6 of the 71 cefotaxime-resistant strains, 5 of the 38 gentamicin-resistant strains and 7 of the 43 ceftazidime-resistant strains, with a major error rate of 0.001 or 0.002 for each group. That combination is the hallmark of a classifier that has collapsed onto the majority class: it hardly ever makes a resistant prediction, so it hardly ever makes such predictions wrongly. These numbers are more difficult to dispute than the PR-AUC values above since VME and ME are calculated within a class and stay fixed as the class mix varies. Prevalence-matching resamples only susceptible strains, and very major error rate is considered to be all in the resistant category, thus cannot be resampled. Clinical deterioration here therefore has no prevalence component, and requires no correction before it can be believed.

TABLE XVII
THE LINEAGE HOLDOUT WITH THE PREVALENCE EFFECT REMOVED. THE PREVALENCE-MATCHED COLUMN RESAMPLES THE HOLDOUT TEST SET TO THE TRAINING PREVALENCE; THE FINAL COLUMN IS THE RESULTING GAP AGAINST THE RANDOM SPLIT.

Antibiotic	Random split	Holdout, as measured	Holdout, prevalence-matched	ROC-AUC	Adjusted gap
AMP	0.9440	0.9043	0.9268 [0.918, 0.938]	0.9257	**−0.017**
CIP	0.9773	0.9118	0.9471 [0.926, 0.967]	0.9739	−0.030
SXT	0.8352	0.7872	0.8046 [0.792, 0.823]	0.9038	−0.031
CTX	0.9019	0.6073	0.7260 [0.687, 0.766]	0.8838	−0.176
CAZ	0.8299	0.4930	0.6431 [0.600, 0.679]	0.7878	−0.187
AMC	0.7482	0.4865	0.5513 [0.529, 0.584]	0.6894	−0.197
GEN	0.7037	0.2457	0.4594 [0.398, 0.511]	0.7426	**−0.244**
Average	**0.8486**	**0.6337**	**0.7226**	**0.8439**	**−0.126**

Note: Once prevalence is corrected for, ampicillin's true gap is −0.017, near nothing, against gentamicin's −0.244; the average true gap of −0.126 is still well below the raw 0.215 gap the uncorrected numbers would suggest. Bold marks the extreme values in a column, best and worst alike.

TABLE XVIII
CLINICAL ERROR RATES UNDER THE LINEAGE HOLDOUT AGAINST THE RANDOM SPLIT, AT A THRESHOLD OF 0.5. BEING WITHIN-CLASS RATIOS, VME AND ME ARE UNAFFECTED BY THE PREVALENCE SHIFT.

Antibiotic	Resistant in test	VME random	VME holdout	Increase	ME holdout	CA random	CA holdout
CIP	132	0.086	0.114	+0.028	0.006	0.969	0.978
AMP	300	0.109	0.160	+0.051	0.095	0.893	0.877
SXT	142	0.236	0.310	+0.074	0.072	0.871	0.863
AMC	145	0.394	0.710	+0.316	0.076	0.804	0.754
GEN	38	0.436	**0.868**	+0.432	0.002	0.934	0.962
CAZ	43	0.291	**0.837**	+0.546	0.001	0.936	0.959
CTX	71	0.190	**0.915**	+0.725	0.001	0.953	0.918
Mean	—	**0.249**	**0.559**	**+0.310**	—	—	—

Note: Gentamicin (0.868), ceftazidime (0.837) and cefotaxime (0.915) all have a holdout VME above 0.83, meaning the model has effectively stopped finding resistance for these three; the mean row averages across all seven.

The categorical agreement column demonstrates why accuracy does not drive this research. Three antibiotics show equal or higher categorical agreement in the holdout than in the random split, and for the other four the reduction in agreement is much smaller than the total collapse in clinical safety alongside it. Gentamicin increases from 0.934 to 0.962 while very major error rate doubles to 0.868; ceftazidime from 0.936 to 0.959, missing 36 of the 43 resistant strains; cefotaxime changes by just 0.035 but its very major error rate rises from 0.190 to 0.915. By the measure a machine learning paper would use to report results, this model looks better or nearly as good, on unseen lineages; by the measure a clinician would use, it became unusable.

This is the third analysis to produce the same grouping. They are the same three with the same saved predictions and test set, therefore they are complementary views but measure different aspects of those predictions. The three antibiotics with negligible loss in PR-AUC gain between 0.028 and 0.074 in very major error rate, and the four with a significant loss have a gain of 0.316 to 0.725. The order within the second group changes, as cefotaxime is worst in this group whereas gentamicin was worst using PR-AUC, the rank within the groups is not to be overly read, but the grouping does not change across all three analyses.

G. Component Performance Under the Lineage Holdout

The holdout also allows us to redo the component analysis of Section V-C in more difficult conditions, and Table XIX shows that the outcome alters the previous picture.

TABLE XIX
AVERAGE PR-AUC OF EACH COMPONENT OF THE MODEL, UNDER THE LINEAGE HOLDOUT.

Component	Avg PR-AUC
TriLens neural arm alone (lenses with gate)	0.472
Five classical learners alone	**0.628**
Full hybrid model	**0.634**

Note: The classical learners alone reach 0.628, close to the full hybrid model's 0.634 under this harder split, meaning the neural lenses add almost nothing here. The 0.634 in the last row is the 0.6337 reported to four decimal places in Table XVI. The 0.628 in the middle row is the five base learners *stacked*, and is a different quantity from the 0.6299 of Table XIV, which is the single strongest learner on its own: under this split the stacking layer recovers slightly less than its own best member already had.

The neural arm alone, that is the three lenses with the gate, scored 0.781 on the random split. It plummets to 0.472 here, when tested on truly unseen lineages; this represented a much larger decline compared to what the classical learners achieve on their own (0.628), nearly matching the full hybrid model (0.634). The classical learners do nearly all of the work under a true test of generalisation and the neural lenses add almost nothing. It is a stronger form of what Section V-C already indicated: the gap between the two arms is 0.156, and the fused model only recovers 0.006 of it. But this was not a case where some other deep learning design would have generalised better; TabNet reran on this split averages 0.326, worse than our neural lenses on their own and well below the classical

TABLE XX
THE CLASSICAL STACKING HALF OF THE PIPELINE, TRAINED INDEPENDENTLY ON A SECOND COLLECTION. NEITHER COLUMN IS THE FULL TRILENS MODEL.

Antibiotic	Arcadia	Giessen	Retention
CIP	0.9773	0.9868	**101.0%**
CAZ	0.8299	0.7775	93.7%
CTX	0.9019	0.7876	87.3%
GEN	0.7037	0.5480	**77.9%**
Average	**0.8532**	**0.7750**	**90.0%**

Note: Ciprofloxacin's apparent 101.0% retention is the highest in the table and gentamicin's 77.9% the lowest, matching the ordering seen under the lineage holdout. As the text below explains, these retention figures are inflated by Giessen's higher prevalence rather than by genuinely better performance. The average retention of 90.0% is the mean of the four per-drug ratios; the ratio of the two column means is 90.8%. Bold marks the extreme values in a column, best and worst alike.

learners. Gradient-boosted trees on gene-level features are the best method we identified on this dataset and under this type of test.

H. Transfer to an Independent Collection

The downside on unseen lineages gives one an obvious next question: does it work at all on a different set of strains? The two datasets do not have a shared feature vector as Arcadia uses the pan-genome reference with 535,000 variant positions and Giessen uses a single reference with approximately 60,000 variant positions. We rather trained the classical stacking part of the pipeline on Giessen with the same protocol and hyper-parameters. The three neural lenses require feature shapes only the pan-genome view can provide and thus are not included in this comparison; and four antibiotics are in common to both collections.

The approach keeps about 90% of its performance on a genomically different collection. It is not zero-shot transfer, in that the model was retrained; what it demonstrates is that our design decisions are not optimized to the specifics of a single dataset.

There is one caveat here, the same one this section raises so many times elsewhere. Section V-E holds that PR-AUC cannot be compared across partitions with differing prevalence, and the two collections differ significantly in prevalence. In Arcadia, the resistance of ciprofloxacin stands at 0.228 as compared to 0.452 in Giessen. Cefotaxime is 0.152 resistant in Arcadia as compared to 0.443 in Giessen. In Arcadia, ceftazidime is 0.131 resistant against 0.341 in Giessen. Gentamicin has a resistance of 0.118 in Arcadia compared to 0.232 in Giessen. In Giessen, every drug's no-skill baseline is about twice to three times Arcadia's, before the model is credited with any skill at all. So the retention percentages are misleading by this difference in baselines, not by any actual improvement, and perhaps the most transparent example is ciprofloxacin whose apparent 101% indicates that the model is marginally better on Giessen, but what really happened is that Giessen made the task marginally easier. The bias cannot be eradicated, because prevalence-matching would require both collections to be resampled to an identical reference, and the 81-strain Giessen test partition is too small to survive that exercise. The weaker argument is the one we will justify from this table: that the recipe reconstructs and does not fall apart, not that performance is maintained to within ten percent of its original extent. The interesting part is the per-antibiotic pattern. Ciprofloxacin retains most of its performance and gentamicin only 77.9% of its own, the same order as under the lineage holdout, though reached by an entirely different route.

I. Marker Genes and Their Annotation

We obtained the most impactful genes based on two sources: the weights of attention within the three neural lenses, and SHAP values obtained with the five classical base learners. This provided 1,138 marker genes in total which covered all antibiotics. Their Prokka [47] locus tags alone do not provide us with any information, so we aligned them with the eggNOG [48] functional annotation released with the Arcadia dataset. We were successful in annotating 889 of the 1,138 markers, or 78.1%, with the others having no corresponding match and indicating them as not annotated instead of making a best guess. Eight of the annotated markers were known genes involved in antimicrobial resistance: *gyrA*, *parC*, the two target genes where point mutations result in fluoroquinolone resistance; *emrB*, *mdtC*, *mdtD* and *mdtO*, which are parts of multidrug efflux pumps; *cusA*, which belongs to the copper and silver efflux system and is therefore a heavy-metal rather than a drug transporter, so we count it as an efflux marker but not as direct evidence of an antibiotic mechanism; and one additional gene annotated as a beta-lactamase.

The genuine result is in finding *gyrA* and *parC* and it is more significant following the lineage holdout than ever. These are the textbook objectives of ciprofloxacin resistance and ciprofloxacin was the very antibiotic that did not succumb to the holdout most, dropping only 0.030 when prevalence is held constant, as compared to other antibiotics that dropped up to 0.244. A model based on an actual, conceptually clear mutation should be transferred to a new lineage and that is what we observe.

A single detail is more important than the list of genes. The eight resistance genes were all the result of the SHAP analysis of the classical learners. None was a result of attention weights of the three neural lenses, although 420 out of the 1,138 markers were selected out of those lenses.

Two things besides a real failure could bring about that zero, and we tested each of them. The first is annotation coverage, as an eggNOG file lacks a gene and there is no way to identify it as a resistance gene no matter how significant the model considered it to be. The two of three lenses are eliminated by coverage: each of 140 markers contained in the Cross-Antibiotic Context Module was annotated, and 97.9% of markers in the Motif Pattern Encoder, but neither generated a single resistance gene. The weaker case is the Gene Feature Attention Module at 61.4%, which is important as it is the lens the gate is most dependent on, thus the view that our model believes is best will be the one that we can reasonably interpret least. The second option is that twenty genes per lens per drug is too small a window. We therefore repeated the annotation

on the top 100 genes instead of the top 20, a fivefold widening of the window, which raised coverage to 79.3%. It also did not recover any resistance gene, so the zero is not an artefact of the depth at which the ranked list was truncated.

Another measurement can be used to explain why. The variant sweep that followed the final lens design concentrated focus on the 5,000 selected genes instead of the entire catalogue of 32,441 genes, thus the maximum attainable entropy there was $\ln 5{,}000 \approx 8.52$, not the $\ln 32{,}441 \approx 10.39$ of the full set. Compared against this lower ceiling, attention entropy remains at 8.51 for trimethoprim-sulfamethoxazole, effectively at ceiling, and above 7.79 for amoxicillin-clavulanate, gentamicin, cefotaxime and ceftazidime, still at or within ten percent of it. When the attention mass of the five drugs is at the most not even one percent of the total mass, in any case, and when ninety percent of it is distributed amongst two or four thousand genes, the pooling layer has virtually broken down into a mere mean. Only ampicillin and ciprofloxacin exhibit real concentration, with entropies 5.34 and 5.46 with just under twenty genes exceeding one percent each, and we by no means attribute biology to even that, since another pooling function tested in a separate sweep concentrated on another group of drugs, indicating the pattern follows the architecture and not the resistance mechanism.

What it does focus on is still informative. Each gene in the pan genome has a locus tag that indicates which of the assemblies the gene was called from, providing an inexpensive method to determine which of the attended genes are sampled or from across the pan genome. For the Gene Feature Attention Module they are not: for each of the seven antibiotics, the top genes by this module were concentrated in a single assembly between 12 times that assembly's fraction in the catalogue and 23 times: median was 16.4. Those counts are less informative because they represent a small number of genes per antibiotic, only genes with an assembly tag can be counted and that counts less than a dozen genes in total, so we read the ratio more as an indicator of the direction of the lens than as a formal test.

An obvious objection is that this is simply an artefact of how pan-genome clusters are named, as each cluster takes the tag of whichever assembly represented it, and a well represented assembly would then swamp any list by arithmetic. There are two reasons why not to think so. The assembly the lens focuses on provides merely 3.3% of the catalogue, therefore, abundance can not account for a twenty-fold preference. And the other two lenses do only just what the artefact explanation would predict: both focus on the *most* populous tag they are reading, at an enrichment of only 1.6 to 2.1, so that we do not report them as a discovery.

J. Ablation of Strain Conditioning in Lens B

These above are what attention still cannot do. They are, however, much better than the results obtained via a direct application of gated attention pooling on this input, and this difference is one of the design contributions of this work: it is the only area where we changed the mechanism itself rather than the data it reads from. Section IV-C2 and Equations (12)–(13) set out the argument in full: on a presence-and-absence vector the standard gated-attention score becomes simply that gene's embedding, and is thus identical for all strains that carry that gene, and conditioning it on the mean embedding of the genes the strain actually carries restores per-sample behaviour. As this reasoning predicts a behaviour, we evaluated it in an ablation, training the model in both directions using the same seeds, folds, splits and hyperparameters, changing only this single term. Table XXI reports the effect.

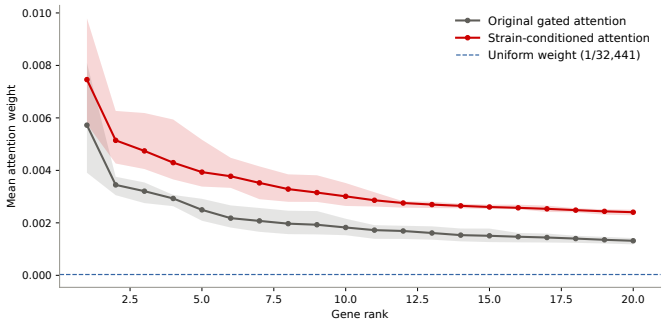


Fig. 11. Mean attention weight against gene rank in the Gene Feature Attention Module, averaged over the seven antibiotics, before and after strain conditioning. The shaded band is the full minimum-to-maximum range across the seven antibiotics at each rank, and the dashed line is the weight a uniform distribution would place on every one of the 32,441 genes.

Three of the four interpretability measures go in the expected direction, and one does not. As Fig. 11 shows, the strain-conditioned lens concentrates more weight on its highest-ranked genes at every rank, and the resistance information of the attended genes more than doubles from 0.0192 to 0.0419, which is 89% of the SHAP-selected genes on average, and greater than them on five out of seven antibiotics. The focus on a single assembly drops from 16.4-fold to 9.0-fold and three out of the seven drugs are no longer significant at the 5% threshold, and the number of independent source assemblies among the attended genes almost doubles, so the provenance signature is mostly diluted.

The exception is U_{ST}, which rises from 0.3123 to 0.3569. By the convention of Fig. 12, where a lower value means a gene set is less determined by lineage, that is a movement in the wrong direction, and we report it as such rather than presenting the ablation as uniformly favourable. Both values remain well below the 0.5390 of a random gene, so the attended set is still less lineage-determined than the pan-genome at large, but strain conditioning buys its extra resistance information at the cost of some of that margin. The two quantities are not independent: genes that carry resistance information in this collection are also, on average, more lineage-structured, which is the same coupling that Section VI identifies as the reason four of the seven antibiotics fail to transfer. We therefore read the pair together, as a trade rather than as a clean improvement.

The rows of performance are not as plain as they appear at the first sight as the term buys practically nothing where it can be measured with ease and much where it is needed. Strain conditioning is worth almost nothing where it is easiest to measure: $+0.0024$ on the random split, and $+0.0037$ on the full stack under the lineage holdout. It is worth an order of

TABLE XXI
ABLATION OF STRAIN CONDITIONING IN THE GENE FEATURE ATTENTION MODULE. EVERYTHING BUT THIS ONE TERM IS IDENTICAL, AND THE "FIXED" COLUMN IS THE DELIVERED MODEL REPORTED EVERYWHERE ELSE. THE REFERENCE COLUMN GIVES THE COMPARISON VALUE THE ROW IS READ AGAINST, AND IS NOT PART OF THE ABLATION.

Measure	Standard	Fixed	Change	Reference
Predictive performance				
Random split, full stack	0.8462	0.8486	+0.0024	—
Phylogroup-stratified split	0.8519	0.8531	+0.0012	—
Lineage holdout, full stack	0.6300	0.6337	+0.0037	—
Lineage holdout, classical alone	0.628	0.628	±0	—
Holdout, neural lenses alone	**0.437**	**0.472**	**+0.035**	—
Clinical safety and interpretability				
Mean very major error rate	0.257	0.249	−0.008	—
Resistance genes by attention	0	**1**	+1	SHAP 8
U_{res} of attended genes	0.0192	**0.0419**	+0.0227	SHAP 0.0473
U_{ST} of attended genes	0.3123	0.3569	+0.0446	random 0.5390
Enrichment on one assembly	16.4×	9.0×	−7.4×	—

Note: Under the lineage holdout, the neural lenses alone gain 0.035 from strain-conditioning, an order of magnitude more than the 0.0024 it buys on the random split; the two interpretability rows move with it, from zero resistance genes recovered to one, and U_{res} nearly doubling from 0.0192 to 0.0419. U_{ST} is the one measure that moves the wrong way, as the text below discusses.

magnitude more, +0.035, on the neural lenses alone under that same holdout, while the classical learners do not move at all and stay at 0.628. An ablation run only on the random split would have hidden this completely. The degenerate scoring function was costing the neural arm most of what it lost anywhere, on precisely the split this research is interested in, and the classical arm was strong enough that none of that loss ever showed in the headline number.

Our most interesting change is the one which we have not tuned for. The attention restores MFCAOJAD_04226, a beta-lactamase, with the score conditioned to the presence of the strain, and selectively recovers it with ampicillin and amoxicillin-clavulanate, the two penicillins in our panel and the two drugs which a beta-lactamase can explain being resistant to. It is the locus tag that SHAP had already determined in the classical learners and thus, two attribution paths that previously disagreed on all points now converge on a single gene, independently. We do not exaggerate this: it is a single gene per hundred genes tested each lens per drug, it is ranked 19 and 20 instead of first, and *gyrA* and *parC* have not appeared in the attention rankings.

One boundary applies. This is information on the presence and absence vector for attention, and is specific to that input, not to attention pooling in general. What generalises is that the diagnostic is an attention mechanism whose scores do not change across samples, which means that we cannot obtain a per-sample explanation and we can figure out that fact in minutes, without retraining anything. We computed the fraction of presence-absence entropy, $U_{ST}(g) = I(g;\mathrm{ST})/H(g)$, which is equal to 1 if a gene's presence is completely attributed to its lineage and 0 if it says nothing about the gene's lineage. In Table XXII the selected genes are compared with the genes attended and to 500 random selections of equal size.

There are two things that follow, both depicted in Fig. 12. The first is not a feature of our model itself, but of the data: the presence-absence pattern of a random gene from this pangenome is explained to 54% by sequence type. Such strongly lineage-structured content of *E. coli* accessory genes is the most obvious single reason why a lineage-sensitive split was required in the first place.

TABLE XXII
HOW MUCH OF EACH GENE SET IS EXPLAINED BY LINEAGE (U_{ST}) AND BY RESISTANCE (U_{res}), OVER THE TOP TWENTY GENES PER ANTIBIOTIC. THE ATTENTION ROW IS THE STANDARD FORMULATION; THE DELIVERED STRAIN-CONDITIONED LENS REACHES 0.357 AND 0.0419.

Gene set	U_{ST}	vs. rand.	U_{res}	vs. attn.
Random genes (500 draws)	0.539	N/A	N/A	N/A
SHAP, classical learners	0.375	0.70×	**0.0473**	2.5×
Attention, Lens B	**0.312**	0.58×	**0.0192**	N/A

Note: Attention's U_{ST} of 0.312 is lower than a random gene's 0.539, so the attended genes are not a lineage marker; its U_{res} of 0.0192 is less than half of SHAP's 0.0473, so they carry far less resistance information than the genes the classical learners select.

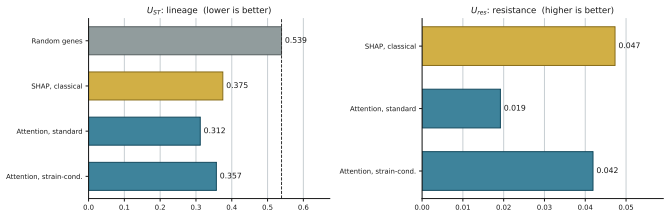


Fig. 12. The attended gene sets against SHAP and against random draws, on both measures. Lower is better on the left, higher is better on the right.

The second one is that the genes attended are not a lineage marker. They are *less* lineage-determined by the standard formulation, at 0.312; and by the delivered formulation, at 0.357, than a random gene; and nothing in the 500 random draws was below them. Nor do they hold much information of resistance: their U_{res} of 0.0192 is the lowest of the three sets, about half the 0.0473 that the genes SHAP chooses do, and even the strain-conditioned version is only 0.0419. It is not the lens that read the wrong thing but the lens that read hardly anything. The useful part is the comparison with SHAP

in that both sets of genes were selected on the same trained pipeline on the same features: the classical side picks genes with two and a half times more resistance information. This matches the result reported in Section V-E, where the neural lenses alone under the lineage holdout remain at 0.472 while the classical learners remain at 0.628. It is the same learners who generalise that are also the ones which pick the more informative genes.

This conflict does not exist solely on the neural side. In the classical arm, the two attribution modes also struggle to agree to a great degree. When the top twenty genes ranked by XGBoost gain are compared with the top twenty genes ranked by SHAP, by each antibiotic and by both sets of classical features, the Jaccard overlap ranges between 0.053 and 0.212, with an average of 0.11 on the gene-aggregated features and 0.17 on the raw SNP features, so at best only four to six genes out of twenty will be shared between the two lists. The rank correlation is not so good: Spearman is negative with all seven antibiotics, between -0.04 and -0.53. That figure needs a qualification before it is read as a disagreement about the model. It is computed over the union of the two top-twenty lists, not over the full ranking of 5,000 genes, so it describes only how the two methods order the handful of genes that at least one of them already considers important. Over so short and so truncated a range a correlation is unstable and its sign carries little weight; what the negative values establish is the absence of agreement at the top of the ranking, not the presence of a systematic inversion. The Jaccard figures are the more reliable of the two statistics for the same reason. Even on the weaker reading, the results of two methods of attributing to the same trained model are not expected to overlap this little, and hence neither can be taken as the interpretation of the model itself. The findings of the gene level, then, are reported as leads but not settled biology, such as the eight resistance genes above, which are informative because they are all in agreement with independently determined biology but not because the attribution method has ascertained them.

VI. Discussion

A. Mechanism Against Population Structure

With the corrected split in hand, the pattern across antibiotics becomes much clearer than it was before. The most obvious example of a learnt real mechanism is ciprofloxacin. It also decays with a 0.030 loss when prevalence is held fixed and its two known resistance genes are in our annotated markers. *E. coli* resistance to fluoroquinolone is typically due to point mutations in these genes: a small, precisely defined signal, and precisely that which a model can learn once, and then recall in an unrelated isolate.

Gentamicin is the opposite case. When prevalence is fixed it still loses 0.244, the largest gap of the seven, and no aminoglycoside-resistance gene feature enters into the final model. The gentamicin resistance often comes on plasmid and mobile element genes and is transferred in a lineage specific manner far more than it is across lineages, and does not assemble well in a pan-genome built this way with this dataset. The model has learned essentially which lineage a strain belongs to, not which gene it harbors. Lastly, ceftazidime and cefotaxime, which have losses of 0.187 and 0.176, respectively, potentially share overlapping mobile beta-lactamase elements.

This account is not fully separable from a decision made in our own pipeline, and we will not present it as though it were. Section III-D sets missing genotype calls to the reference allele, following the authoring group, and notes there that a call fails most often on plasmids and other mobile elements that assemble poorly from short reads. That is the same set of loci on which the four difficult antibiotics depend. The biological reading above and the artefact of our imputation therefore predict the identical outcome on the identical drugs, and nothing in the experiments reported here distinguishes them. Two observations keep the biological reading in play: amoxicillin-clavulanate, whose usual mechanism is a change in expression level rather than the acquisition of a gene, fails in the same group despite not depending on assembly at all; and the ordering of the seven drugs reproduces under the independent Giessen collection, which was called against a different reference. Neither is decisive. Repeating the pipeline with missing calls masked rather than imputed would separate the two, costs nothing but compute, and is listed in Section VII as future work. Until it is done, the mechanism account should be read as the more probable of two live explanations rather than as an established one.

The most obvious example of another failure is amoxicillin-clavulanate that loses 0.197. The expression of more enzyme or release of a membrane channel, changes in degree, rather than changes in which genes are expressed, are often the basis of its resistance. This type of resistance can never be easily overcome by our presence-and-absence lens, which bears most of the weight of the entire gate during the course of this research, and therefore this type of resistance was never going to be easy to capture in our design.

All this does not make the model useless. It is efficient on simple, well known mechanisms and inefficient on resistance as carried by mobile, plasmid-borne elements or by low level changes in gene expression. This is a specific, falsifiable claim about the current state of genomic AMR prediction, and a more useful conclusion than an average score would have been.

B. Answers to the Research Questions

Taking the five research questions of Section I in turn.

RQ1. TriLens-MoE-Stack achieved an average PR-AUC of 0.849 on the random split, the highest in Table XIV. It is ahead of every external baseline and ahead of its strongest component on five of the seven antibiotics. But its margin over the best single classical learner is only 0.009, which falls within the bootstrap intervals and is of the size that seed variation alone could produce, since every model here was trained once (Section V-B). The claim that can be defended is therefore the weaker one: nothing simpler beats it, rather than that it beats everything simpler.

RQ2. Each component has its contribution to make, and the classical arm has the greatest: the addition of the gate is 0.035

and the classical step of stacking 0.068 to the random split (Section V-C). According to the lineage holdout, the classical learners are the only ones to hit 0.628, compared to the entire model 0.634, and the neural lenses are the only ones to hit 0.472, so on truly new lineages, the neural side contributes almost nothing. The 0.068 that stacking adds is comfortably larger than the interval widths of Section V-B; the 0.035 that the gate adds is not, and is reported as a direction rather than as a quantity.

RQ3. Answered in the negative but a diagnosis was made. The gate weights can be read but are inconsistent with the prediction of the design, diverting ciprofloxacin off of the motif lens designed to receive it. The attention weights reclaim a single known resistance gene in 420 markers, eight in SHAP on the classical arm. Section V-J shows that this failure can be attributed to a particular and remediable attribute of the lens, an attribute more valuable than a positive answer we cannot justify.

RQ4. A blend with relative proportions that change dramatically by antibiotic. Coarse phylogroup membership has virtually no signal on its own (0.291), and gene annotation recovers *gyrA* and *parC* unprompted, but the prevalence-corrected holdout indicates that finescale relatedness has a lot: three drugs lose 0.017 to 0.031 and four lose 0.176 to 0.244, with no results in between. There is no one answer to RQ4, just a per-antibiotic one.

RQ5. Partially incomplete and uneven generalisation. Average performance drops from 0.849 to 0.634 (or 0.723 after correcting for the prevalence shift) and leftover duplicates are ruled out by a near-duplicate scan. It holds reasonably well for antibiotics with a simple well-understood mechanism and badly for antibiotics whose resistance is linked to mobile genetic elements or to changes in gene expression.

VII. Conclusion

A genomic resistance model can look excellent on paper and still be measuring the wrong thing. That doubt is where this research started. The question we set out to answer was how much of the score comes from real biological patterns, and how much comes from similarities between related strains. We had to have a powerful model in order to test this fairly. TriLens-MoE-Stack was able to obtain an average PR-AUC of 0.849 on the random split, the best score of all the models tested. Against every method from outside this work the lead is clear: 0.045 over the PCA gradient-boosting model this project began with, 0.154 over TabNet, and 0.557 over the null model, and on three of the seven antibiotics the score clears the upper bound of that baseline's confidence interval. Against its own strongest component the margin is 0.009, which is what an ablation is expected to show once the best part is already in place. Then whole sequence types were removed from training and no test strain had a close neighbour in the training set. The average PR-AUC decreased to 0.634. A portion of this decrease was due to a decreased fraction of resistant strains in the holdout set which has a direct impact on PR-AUC. This change was corrected, and the score was 0.723. The actual difference was not 0.215 but 0.126. Also, the decline in performance was quite distinct across antibiotics. Three antibiotics lost only 0.017 to 0.031, while four lost 0.176 to 0.244. This stark division indicates that the resistance associated with the stable biological processes can be transferred to new lineages, whereas the patterns primarily depending on the ancestry are not very transferable. When the classical part of the pipeline was rebuilt on a second dataset, it kept about 90% of its earlier performance. That figure applies to the classical stacking half alone, since the neural lenses need a feature space the second collection does not provide, and Giessen's higher resistance prevalence raises the floor against which PR-AUC is read, so the number is better treated as evidence that the recipe reconstructs than as a measurement of how much accuracy survives. What it indicates is that the design decisions of that half are not tied to the specific characteristics of a single dataset.

However, there are also significant limitations to these results. None of the antibiotics passed both CLSI error limits, so the error rates stay above the level those limits set. That weakness is concentrated in a few drugs rather than spread across the panel. For ciprofloxacin and ampicillin the model reaches 0.9773 and 0.9440 on the random split, and still holds 0.9118 and 0.9043 when every close relative is removed from training. The lineage holdout led to a higher share of resistant strains being incorrectly classified as susceptible (0.249 to 0.559), although the accuracy-like measure did not change significantly. This distinction is important since the accuracy-like measure is commonly reported in the majority of studies and it may conceal severe clinical errors. The results of the interpretation were also not entirely consistent. In the attention weights, only one known resistance gene was identified whereas eight were identified using post-hoc attribution to classical learners. Thus the gene-level results should be viewed as potential leads rather than established biological truths. There is also a definite limit to the gene presence and absence lens. It is able to identify the presence of a gene, but not the level of gene expression. This may account for amoxicillin-clavulanate being the most difficult antibiotic during the study. One more limitation arises from the stratified split that is based on the phylogroup labels. These labels were extracted from an unreferenced lookup table for 57 sequence types and we were not able to verify the original source of that table; calling the phylogroup directly from each genome with an established *in silico* typing tool [37] would put that split on a documented footing.

Four further limitations bound what the numbers above can carry. First, every model was trained once, under a single seed, so the intervals we report cover test-set sampling but not training variability, and the smallest differences in Tables XIV and XXI should not be read as established effects. Second, the lineage-only and PCA baselines were evaluated on the random split but not under the lineage holdout, which is precisely the partition on which a lineage-only baseline would be most informative about the question this work asks. Third, the comparison set contains no rule-based resistance-gene caller of the kind a diagnostic laboratory would actually use, so the external baselines measure this model against other statistical models rather than against current practice. Fourth,

missing genotype calls were set to the reference allele, a choice that biases against exactly the mobile and plasmid-borne loci on which the four difficult antibiotics depend, so the mechanism account in Section VI is not fully separable from that imputation choice. Each of the four is addressable with the data already in hand and none requires a new collection.

The key takeaways from this work are related to testing and refining genomic resistance models. An adequate lineage holdout should eliminate entire clonal groups, but it should also guard against the domination of the test set by a single large clone. This produces another issue as the prevalence of resistance can be reduced with the removal of large resistant groups, which can then lead to a reduction in PR-AUC even if the true ability of the model has not reduced as much. In this study, prevalence correction moved the generalisation gap from 0.215 to 0.126. Thus, approximately two fifths of the original drop was caused by the evaluation measure instead of an actual loss in model performance. A critical issue with gated attention pooling for binary gene data was also uncovered in this study. The original attention score could be the same for all strains, as each present gene used only its fixed embedding. When the score was based on the complete set of genes in each strain, the sample-specific behaviour was regained and the neural part improved by 0.035 PR-AUC under the lineage holdout. The first item of future work is to repeat the whole protocol over several seeds and report a mean and a standard deviation in place of each single figure, since that is what would settle which of the smaller margins here are real. Beyond that, future efforts should be directed towards enhancing detection and assembly of plasmids and other mobile genetic elements, which are likely to harbour missing signals for the four challenging antibiotics, and towards repeating the imputation of missing calls as a masked rather than a reference-allele operation, which would separate the mechanism account from the pipeline choice. Future holdout designs should also maintain a minimum number of resistant strains for each antibiotic, particularly when the results are based upon only 38 cases of resistance (as in the gentamicin score). Finally, the entire study should be performed with another bacterial species to determine if the two-group phenomenon is unique to *Escherichia coli* or is a general pattern for genomic resistance prediction. The main contribution of this work is not a single headline score. It is the finding that genomic resistance prediction in *E. coli* is not one problem but two, and that the boundary between them is sharp, reproducible and specific to each antibiotic. The model is what produced that finding, and the evaluation protocol is what made it measurable.

Data and Code Availability

Both genome collections used here are public. The primary collection is the Arcadia Science *E. coli* genotype and phenotype release [7], [8] and the external collection is the Giessen set of Ren et al. [27]. Every split, filter and model in this report is driven by a single fixed seed (42), so every number reported here is exactly reproducible, but no figure in this paper carries a training-variability estimate, for the reason given in Section V-B. The code, the split definitions and the saved out-of-fold predictions needed to reproduce every number and figure will be released in a public repository on publication.

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